
UNIT – II

FOURIER SERIES

Engineering Mathematics – III

Class Notes
(Digitized)

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1 Introduction to Fourier Series

Definition

Writing the function $f(x)$ in terms of an **infinite sum of sine and cosine terms** is known as the **Fourier series** of $f(x)$.

1.1 Conditions for Fourier Expansion

Dirichlet's Conditions

A function $f(x)$ defined in the interval $(\alpha, \alpha + 2\pi)$ can be expanded in a Fourier series if:

- It must be a **single valued** function.
- It must have a **finite number of maxima and minima**.
- It must have a **finite number of discontinuities**.
- It must be **continuous or piecewise continuous**.

Further, the function must be **periodic**:

$$f(x + 2\pi) = f(x).$$

1.2 Fourier Expansion in the Interval $(0, 2\pi)$

Fourier series in $(0, 2\pi)$

The Fourier expansion of $f(x)$ in the interval $(0, 2\pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

where

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx.$$

1.3 Fourier Expansion in the Interval $(-\pi, \pi)$

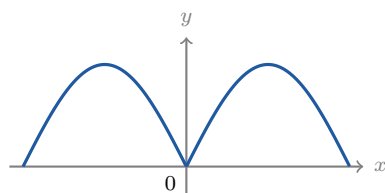
Fourier series in $(-\pi, \pi)$ – length 2π

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

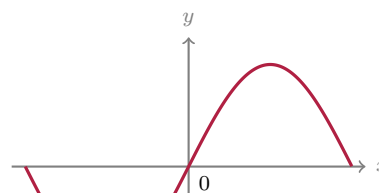
where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx.$$

1.4 Even and Odd Functions



$f(x)$ is **even**
 $f(-x) = f(x)$
 symmetric about the y -axis



$f(x)$ is **odd**
 $f(-x) = -f(x)$
 symmetric about the origin

Case 1 – $f(x)$ is an even function in $(-\pi, \pi)$

Its Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx,$$

where

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad b_n = 0.$$

Case 2 – $f(x)$ is an odd function in $(-\pi, \pi)$

Its Fourier series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx,$$

where

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx, \quad a_0 = 0, \quad a_n = 0.$$

2 Problems on Fourier Series in $(0, 2\pi)$ and $(-\pi, \pi)$

2.1 Problem 1 : $f(x) = (\pi - x)^2$ in $0 \leq x \leq 2\pi$

Problem 1

Find the Fourier series expansion of $f(x) = (\pi - x)^2$, $0 \leq x \leq 2\pi$, and hence show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}.$$

Solution. Given $f(x) = (\pi - x)^2$, $0 \leq x \leq 2\pi$.

Since the interval is $(0, 2\pi)$ of length 2π , its Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Evaluation of a_0

$$\begin{aligned}
a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx \\
&= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 dx \\
&= \frac{1}{\pi} \left[\frac{(\pi - x)^3}{3(-1)} \right]_0^{2\pi} \quad \left(\because \int (\pi - x)^2 dx = \frac{(\pi - x)^3}{3 \cdot (-1)} \right) \\
&= -\frac{1}{3\pi} [(\pi - 2\pi)^3 - (\pi - 0)^3] \\
&= -\frac{1}{3\pi} [(-\pi)^3 - \pi^3] \\
&= -\frac{1}{3\pi} (-\pi^3 - \pi^3) = -\frac{1}{3\pi} (-2\pi^3)
\end{aligned}$$

$$a_0 = \frac{2}{3}\pi^2$$

Evaluation of a_n

Using Bernoulli's rule $\int uv dx = uv_1 - u'v_2 + u''v_3 - \dots$ with $u = (\pi - x)^2$ and $v = \cos nx$:

$$\begin{aligned}
a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \cos nx dx \\
&= \frac{1}{\pi} \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x)(-1) \left(\frac{-\cos nx}{n^2} \right) + 2(-1)(-1) \left(\frac{-\sin nx}{n^3} \right) - 0 \right]_0^{2\pi} \\
&= \frac{1}{\pi} \left[\frac{1}{n} ((\pi - x)^2 \sin nx)_0^{2\pi} - \frac{2}{n^2} ((\pi - x) \cos nx)_0^{2\pi} - \frac{2}{n^3} (\sin nx)_0^{2\pi} \right] \\
&= \frac{1}{\pi} \left[\frac{1}{n} (0 - 0) - \frac{2}{n^2} (-\pi(1) - \pi(1)) - \frac{2}{n^3} (0 - 0) \right] \\
&\quad \left(\because \sin 2n\pi = 0, \cos 2n\pi = 1, \cos 0 = 1 \right)
\end{aligned}$$

$$a_n = \frac{1}{\pi} \left[-\frac{2}{n^2} (-2\pi) \right] = \frac{1}{\pi} \cdot \frac{4\pi}{n^2}$$

$$a_n = \frac{4}{n^2}$$

Evaluation of b_n

With $u = (\pi - x)^2$ and $v = \sin nx$:

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \sin nx \, dx \\
 &= \frac{1}{\pi} \left[(\pi - x)^2 \left(\frac{-\cos nx}{n} \right) - 2(\pi - x)(-1) \left(\frac{-\sin nx}{n^2} \right) + 2(-1)(-1) \left(\frac{\cos nx}{n^3} \right) - 0 \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} \left((\pi - x)^2 \cos nx \right)_0^{2\pi} - \frac{2}{n^2} \left((\pi - x) \sin nx \right)_0^{2\pi} + \frac{2}{n^3} \left(\cos nx \right)_0^{2\pi} \right] \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} \left(\pi^2(1) - \pi^2(1) \right) - \frac{2}{n^2} (0 - 0) + \frac{2}{n^3} (1 - 1) \right] \\
 &= \frac{1}{\pi} [0 - 0 + 0]
 \end{aligned}$$

$$b_n = 0$$

Fourier series

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \\
 &= \frac{\frac{2}{3}\pi^2}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx + 0
 \end{aligned}$$

Required Fourier series

$$(\pi - x)^2 = \frac{\pi^2}{3} + 4 \left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right] \quad (1)$$

Deduction

Put $x = 0$ in the above Fourier series (1):

$$(\pi - 0)^2 = \frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\pi^2 - \frac{\pi^2}{3} = 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\frac{2\pi^2}{3} \times \frac{1}{4} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$\therefore \boxed{\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}}$$

2.2 Problem 2 : $f(x) = x \sin x$ in $-\pi \leq x \leq \pi$ **Problem 2**

Find the Fourier series expansion of $f(x) = x \sin x$ in the interval $-\pi \leq x \leq \pi$, and hence deduce that

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \cdots = \frac{\pi - 2}{4}.$$

Solution. Given $f(x) = x \sin x$, $-\pi \leq x \leq \pi$.

Test for even / odd

$$f(-x) = (-x) \sin(-x) = (-x)(-\sin x) = x \sin x = f(x)$$

$\therefore f(-x) = f(x)$, so $f(x)$ is an **even function**.

Its Fourier series is therefore

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx.$$

Evaluation of a_0

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} \underbrace{x}_u \underbrace{\sin x}_v dx \\ &= \frac{2}{\pi} \left[x(-\cos x) - (1)(-\sin x) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[-x \cos x + \sin x \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[(-\pi \cos \pi + \sin \pi) - (0 + \sin 0) \right] \\ &= \frac{2}{\pi} \left[-\pi(-1) - 0 \right] = \frac{2}{\pi}(\pi) \end{aligned}$$

$$a_0 = 2$$

Evaluation of a_n

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x \underbrace{\sin x}_B \underbrace{\cos nx}_A dx$$

Using the product formula $\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$ with $A = nx$, $B = x$:

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^\pi x \cdot \frac{1}{2} [\sin(nx + x) - \sin(nx - x)] dx \\
 &= \frac{1}{\pi} \int_0^\pi x [\sin(n+1)x - \sin(n-1)x] dx \\
 &= \frac{1}{\pi} \left\{ \int_0^\pi x \sin(n+1)x dx - \int_0^\pi x \sin(n-1)x dx \right\} \\
 &= \frac{1}{\pi} \left\{ \left[x \left(\frac{-\cos(n+1)x}{n+1} \right) - (1) \left(\frac{-\sin(n+1)x}{(n+1)^2} \right) + 0 \right]_0^\pi \right. \\
 &\quad \left. - \left[x \left(\frac{-\cos(n-1)x}{n-1} \right) - (1) \left(\frac{-\sin(n-1)x}{(n-1)^2} \right) + 0 \right]_0^\pi \right\} \\
 &= \frac{1}{\pi} \left\{ -\frac{1}{n+1} (x \cos(n+1)x)_0^\pi + \frac{1}{(n+1)^2} (\sin(n+1)x)_0^\pi \right. \\
 &\quad \left. + \frac{1}{n-1} (x \cos(n-1)x)_0^\pi - \frac{1}{(n-1)^2} (\sin(n-1)x)_0^\pi \right\} \\
 &= \frac{1}{\pi} \left\{ -\frac{1}{n+1} (\pi(-1)^{n+1} - 0) + \frac{1}{(n+1)^2} (0 - 0) \right. \\
 &\quad \left. + \frac{1}{n-1} (\pi(-1)^{n-1} - 0) - \frac{1}{(n-1)^2} (0 - 0) \right\} \\
 &= \frac{-(-1)^{n+1}}{n+1} + \frac{(-1)^{n-1}}{n-1} = \frac{(-1)^n}{n+1} - \frac{(-1)^n}{n-1} \\
 &= (-1)^n \left[\frac{(n-1) - (n+1)}{(n+1)(n-1)} \right] = (-1)^n \left[\frac{-2}{n^2 - 1} \right]
 \end{aligned}$$

$$a_n = \frac{-2(-1)^n}{n^2 - 1}, \quad \text{provided } n \neq 1$$

Evaluation of a_1 (the case $n = 1$)

Since the above expression fails for $n = 1$, we find a_1 separately, by definition:

$$\begin{aligned}
 a_1 &= \frac{2}{\pi} \int_0^\pi f(x) \cos x \, dx = \frac{2}{\pi} \int_0^\pi x \sin x \cos x \, dx \\
 &= \frac{2}{\pi} \int_0^\pi x \frac{\sin 2x}{2} \, dx \quad (\because 2 \sin x \cos x = \sin 2x) \\
 &= \frac{1}{\pi} \int_0^\pi x \sin 2x \, dx \\
 &= \frac{1}{\pi} \left[x \left(\frac{-\cos 2x}{2} \right) - (1) \left(\frac{-\sin 2x}{4} \right) \right]_0^\pi \\
 &= \frac{1}{\pi} \left[-\frac{\pi \cos 2\pi}{2} + \frac{\sin 2\pi}{4} - 0 \right] = \frac{1}{\pi} \left[-\frac{\pi(1)}{2} \right]
 \end{aligned}$$

$$a_1 = -\frac{1}{2}$$

Fourier series

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \\
 &= \frac{2}{2} + a_1 \cos x + \sum_{n=2}^{\infty} a_n \cos nx \\
 &= 1 - \frac{1}{2} \cos x + \sum_{n=2}^{\infty} \frac{-2(-1)^n}{n^2 - 1} \cos nx \\
 &= 1 - \frac{1}{2} \cos x - 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{(n+1)(n-1)} \cos nx
 \end{aligned}$$

Required Fourier series

$$x \sin x = 1 - \frac{1}{2} \cos x - 2 \left[\frac{\cos 2x}{1 \cdot 3} - \frac{\cos 3x}{2 \cdot 4} + \frac{\cos 4x}{3 \cdot 5} - \dots \right] \quad (1)$$

Deduction

Put $x = \frac{\pi}{2}$ in (1):

$$\begin{aligned}\frac{\pi}{2} \sin \frac{\pi}{2} &= 1 - \frac{1}{2} \cos \frac{\pi}{2} - 2 \left[\frac{\cos \pi}{1 \cdot 3} - \frac{\cos \frac{3\pi}{2}}{2 \cdot 4} + \frac{\cos 2\pi}{3 \cdot 5} - \frac{\cos \frac{5\pi}{2}}{4 \cdot 6} + \frac{\cos 3\pi}{5 \cdot 7} - \dots \right] \\ \frac{\pi}{2} &= 1 - 0 - 2 \left[-\frac{1}{1 \cdot 3} - 0 + \frac{1}{3 \cdot 5} - 0 - \frac{1}{5 \cdot 7} - \dots \right] \\ \frac{\pi}{2} &= 1 + 2 \left[\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right] \\ \frac{\pi}{2} - 1 &= 2 \left[\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right] \\ \frac{\pi - 2}{2} \cdot \frac{1}{2} &= \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \\ \therefore \quad \boxed{\frac{\pi - 2}{4} &= \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots}\end{aligned}$$

2.3 Problem 3 : Square-wave function**Problem 3**

Find the Fourier series of

$$f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$$

and hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

Solution. Given $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$

Its Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Evaluation of a_0

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left\{ \int_{-\pi}^0 (-k) dx + \int_0^{\pi} k dx \right\} \\ &= \frac{1}{\pi} \left\{ -k[x]_{-\pi}^0 + k[x]_0^{\pi} \right\} \\ &= \frac{1}{\pi} \left\{ -k[0 - (-\pi)] + k[\pi - 0] \right\} \\ &= \frac{1}{\pi} \left\{ -k\pi + k\pi \right\}\end{aligned}$$

$$\boxed{a_0 = 0}$$

Evaluation of a_n

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \left\{ \int_{-\pi}^0 f(x) \cos nx \, dx + \int_0^{\pi} f(x) \cos nx \, dx \right\} \\
 &= \frac{1}{\pi} \left\{ \int_{-\pi}^0 (-k) \cos nx \, dx + \int_0^{\pi} k \cos nx \, dx \right\} \\
 &= \frac{1}{\pi} \left\{ -k \left[\frac{\sin nx}{n} \right]_{-\pi}^0 + k \left[\frac{\sin nx}{n} \right]_0^{\pi} \right\} \\
 &= \frac{1}{\pi} \left\{ -k \left[\frac{\sin 0}{n} - \frac{\sin(-n\pi)}{n} \right] + k \left[\frac{\sin n\pi}{n} - \frac{\sin 0}{n} \right] \right\} \\
 &= \frac{1}{\pi} \{ -k(0) + k(0) \} \quad (\because \sin n\pi = 0)
 \end{aligned}$$

$$a_n = 0$$

Evaluation of b_n

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \left\{ \int_{-\pi}^0 f(x) \sin nx \, dx + \int_0^{\pi} f(x) \sin nx \, dx \right\} \\
 &= \frac{1}{\pi} \left\{ \int_{-\pi}^0 (-k) \sin nx \, dx + \int_0^{\pi} k \sin nx \, dx \right\} \\
 &= \frac{1}{\pi} \left\{ -k \left[\frac{-\cos nx}{n} \right]_{-\pi}^0 + k \left[\frac{-\cos nx}{n} \right]_0^{\pi} \right\} \\
 &= \frac{1}{\pi} \left\{ -k \left[-\frac{1}{n} + \frac{\cos(-n\pi)}{n} \right] + k \left[-\frac{\cos n\pi}{n} + \frac{1}{n} \right] \right\} \\
 &= \frac{1}{\pi} \left\{ \frac{k}{n} - \frac{k \cos n\pi}{n} - \frac{k \cos n\pi}{n} + \frac{k}{n} \right\} \quad (\because \cos(-n\pi) = \cos n\pi) \\
 &= \frac{2k}{n\pi} (1 - \cos n\pi) = \frac{2k}{n\pi} (1 - (-1)^n)
 \end{aligned}$$

For n **odd**, $\cos n\pi = -1$, so

$$b_n = \frac{1}{\pi} \left[\frac{k}{n} + \frac{k}{n} + \frac{k}{n} + \frac{k}{n} \right] = \frac{4k}{n\pi};$$

for n **even**, $\cos n\pi = 1$ and $b_n = 0$.

$$b_n = \begin{cases} \frac{4k}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

Fourier series

$$f(x) = \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

Required Fourier series

$$f(x) = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots \right) \quad (1)$$

Deduction

Put $x = \frac{\pi}{2}$ in (1). Since $0 < \frac{\pi}{2} < \pi$, we have $f\left(\frac{\pi}{2}\right) = k$:

$$f\left(\frac{\pi}{2}\right) = \frac{4k}{\pi} \left(\sin \frac{\pi}{2} + \frac{1}{3} \sin \frac{3\pi}{2} + \frac{1}{5} \sin \frac{5\pi}{2} + \frac{1}{7} \sin \frac{7\pi}{2} + \cdots \right)$$

$$k = \frac{4k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right)$$

$$1 = \frac{4}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right)$$

$$\therefore \quad \boxed{1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}}$$

2.4 Problem 4 : $f(x) = x^2$ in $-\pi \leq x \leq \pi$ **Problem 4**

Find the Fourier series of $f(x) = x^2$, $-\pi \leq x \leq \pi$, and hence deduce that

$$(i) \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$$

$$(ii) \quad \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots = \frac{\pi^2}{12}$$

$$(iii) \quad \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

Solution. Given $f(x) = x^2$, $-\pi \leq x \leq \pi$.

Test for even / odd

$f(-x) = (-x)^2 = x^2 = f(x)$, hence $f(x)$ is an **even function**.

$\therefore$ Its Fourier series is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$, with $b_n = 0$.

Evaluation of a_0

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi} = \frac{2}{\pi} \cdot \frac{\pi^3}{3}$$

$$\boxed{a_0 = \frac{2\pi^2}{3}}$$

Evaluation of a_n

Bernoulli's rule with $u = x^2$, $v = \cos nx$:

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^\pi x^2 \cos nx \, dx \\
 &= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) - 0 \right]_0^\pi \\
 &= \frac{2}{\pi} \left[\frac{1}{n} (x^2 \sin nx)_0^\pi + \frac{2}{n^2} (x \cos nx)_0^\pi - \frac{2}{n^3} (\sin nx)_0^\pi \right] \\
 &= \frac{2}{\pi} \left[\frac{1}{n} (0 - 0) + \frac{2}{n^2} (\pi(-1)^n - 0) - \frac{2}{n^3} (0 - 0) \right] \\
 &= \frac{2}{\pi} \cdot \frac{2}{n^2} \pi (-1)^n
 \end{aligned}$$

$$a_n = \frac{4}{n^2} (-1)^n$$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx = \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos nx$$

Required Fourier series

$$x^2 = \frac{\pi^2}{3} + 4 \left[-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \frac{\cos 4x}{4^2} - \dots \right] \quad (1)$$

Deductions

(i) Put $x = \pi$ in (1):

$$(x^2)_{x=\pi} = \frac{\pi^2}{3} + 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) \quad (\because \cos n\pi = (-1)^n)$$

$$\pi^2 - \frac{\pi^2}{3} = 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\frac{2\pi^2}{3} \times \frac{1}{4} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$\therefore \boxed{\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots} \quad (i)$$

(ii) Put $x = 0$ in (1):

$$(x^2)_{x=0} = \frac{\pi^2}{3} + 4 \left(-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right)$$

$$-\frac{\pi^2}{3} = -4 \left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \right)$$

$$\frac{\pi^2}{3} \times \frac{1}{4} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

$$\therefore \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \quad (\text{ii})$$

(iii) Adding (i) and (ii):

$$\begin{aligned} \frac{\pi^2}{6} + \frac{\pi^2}{12} &= 2 \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \\ \frac{2\pi^2 + \pi^2}{12} &= \frac{3\pi^2}{12} = \frac{\pi^2}{4} = 2 \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \\ \therefore \frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \quad (\text{iii}) \end{aligned}$$

3 Fourier Series at a Point of Discontinuity

Value of the series at a point of discontinuity

Suppose the given function $f(x)$ is defined by

$$f(x) = \begin{cases} \varphi(x), & \alpha \leq x < c \\ \psi(x), & c < x \leq \alpha + 2\pi \end{cases}$$

Here $x = c$ is a **point of discontinuity**. At this point the Fourier series of $f(x)$ converges to the *average of the left-hand and right-hand limits*:

$$(f(x))_{x=c} = \frac{(f(x))_{c^+} + (f(x))_{c^-}}{2} = \frac{(\psi(x))_{x=c} + (\varphi(x))_{x=c}}{2}$$

$$(f(x))_{x=c} = \frac{\psi(c) + \varphi(c)}{2}$$

3.1 Problem 1 : $f(x) = 1 - x$ and $1 + x$ on $(-\pi, \pi)$

Problem 1

Find the Fourier series of

$$f(x) = \begin{cases} 1 - x, & -\pi \leq x < 0 \\ 1 + x, & 0 \leq x \leq \pi \end{cases}$$

and hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.

Solution. Given $f(x) = \begin{cases} 1 - x, & -\pi \leq x < 0 \\ 1 + x, & 0 \leq x \leq \pi \end{cases}$

To check even or odd

Replacing x by $-x$:

$$\begin{aligned} f(-x) &= \begin{cases} 1+x, & -\pi \leq -x < 0 \\ 1-x, & 0 \leq -x \leq \pi \end{cases} = \begin{cases} 1+x, & \pi \geq x > 0 \\ 1-x, & 0 \geq x \geq -\pi \end{cases} \\ &= \begin{cases} 1+x, & 0 < x \leq \pi \\ 1-x, & -\pi \leq x \leq 0 \end{cases} = f(x) \end{aligned}$$

$\therefore f(-x) = f(x)$, so $f(x)$ is an **even function**.

Its Fourier series is therefore $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$, with $b_n = 0$.

Evaluation of a_0

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (1+x) dx = \frac{2}{\pi} \left(\frac{(1+x)^2}{2} \right)_0^{\pi} \\ &= \frac{2}{\pi} \left(\frac{(1+\pi)^2}{2} - \frac{1}{2} \right) = \frac{2}{\pi} \left[\frac{1}{2} + \frac{\pi^2}{2} + \pi - \frac{1}{2} \right] \\ &= \frac{2}{\pi} \left[\frac{\pi^2 + 2\pi}{2} \right] = \frac{2}{\pi} \left[\frac{(\pi+2)\pi}{2} \right] \end{aligned}$$

$$a_0 = \pi + 2$$

Evaluation of a_n

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (1+x) \cos nx dx \\ &= \frac{2}{\pi} \left[(1+x) \left(\frac{\sin nx}{n} \right) - 1 \cdot \left(\frac{-\cos nx}{n^2} \right) + 0 \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\frac{1}{n} \left((1+x) \sin nx \right)_0^{\pi} + \frac{1}{n^2} \left(\cos nx \right)_0^{\pi} \right] \\ &= \frac{2}{\pi} \left[\frac{1}{n} (0 - 0) + \frac{1}{n^2} ((-1)^n - 1) \right] \\ &= \frac{2}{\pi n^2} ((-1)^n - 1) \end{aligned}$$

$$a_n = \begin{cases} \frac{-4}{n^2\pi}, & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases}$$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx = \frac{\pi+2}{2} + \sum_{n=1,3,5,\dots}^{\infty} \frac{-4}{n^2\pi} \cos nx$$

Required Fourier series

$$f(x) = \frac{\pi+2}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

Deduction

Put $x = 0$ in the above Fourier series:

$$(f(x))_{x=0} = \frac{\pi+2}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \quad (2)$$

But $x = 0$ is a **point of discontinuity**, so

$$(f(x))_{x=0} = \frac{f(0^+) + f(0^-)}{2} = \frac{(1+x)_{x=0} + (1-x)_{x=0}}{2} = \frac{(1+0) + (1-0)}{2} = 1.$$

Substituting $(f(x))_{x=0} = 1$ in (2):

$$\begin{aligned} 1 &= \frac{\pi+2}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \\ 1 - \frac{\pi+2}{2} &= -\frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \\ \frac{\pi}{2} \times \frac{\pi}{4} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \end{aligned}$$

$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

3.2 Problem 2 : $f(x) = |\cos x|$ in $-\pi < x < \pi$

Problem 2

Find the Fourier series expansion of $f(x) = |\cos x|$ in the interval $-\pi < x < \pi$.

Solution. Given $f(x) = |\cos x|$ in $-\pi < x < \pi$.

Clearly $f(-x) = |\cos(-x)| = |\cos x| = f(x)$, so $f(x)$ is an **even function**.

Note also that on $(0, \frac{\pi}{2})$, $\cos x > 0$ so $|\cos x| = \cos x$, while on $(\frac{\pi}{2}, \pi)$, $\cos x < 0$ so $|\cos x| = -\cos x$.

Evaluation of a_0

$$\begin{aligned}
 a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} |\cos x| dx \\
 &= \frac{2}{\pi} \left\{ \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} (-\cos x) dx \right\} \\
 &= \frac{2}{\pi} \left\{ (\sin x)_0^{\pi/2} - (\sin x)_{\pi/2}^{\pi} \right\} \\
 &= \frac{2}{\pi} \left\{ (1 - 0) - (0 - 1) \right\} = \frac{2}{\pi} (2)
 \end{aligned}$$

$$a_0 = \frac{4}{\pi}$$

Evaluation of a_n (for $n \neq 1$)

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \\
 &= \frac{2}{\pi} \left[\int_0^{\pi/2} \underbrace{\cos x}_B \underbrace{\cos nx}_A dx - \int_{\pi/2}^{\pi} \cos x \cos nx dx \right]
 \end{aligned}$$

Using $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$ with $A = nx$, $B = x$:

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \left[\frac{1}{2} \int_0^{\pi/2} (\cos(nx+x) + \cos(nx-x)) dx - \frac{1}{2} \int_{\pi/2}^{\pi} (\cos(nx+x) + \cos(nx-x)) dx \right] \\
 &= \frac{1}{\pi} \left\{ \int_0^{\pi/2} \cos(n+1)x dx + \int_0^{\pi/2} \cos(n-1)x dx - \int_{\pi/2}^{\pi} \cos(n+1)x dx - \int_{\pi/2}^{\pi} \cos(n-1)x dx \right\} \\
 &= \frac{1}{\pi} \left\{ \left(\frac{\sin(n+1)x}{n+1} \right)_0^{\pi/2} + \left(\frac{\sin(n-1)x}{n-1} \right)_0^{\pi/2} - \left(\frac{\sin(n+1)x}{n+1} \right)_{\pi/2}^{\pi} - \left(\frac{\sin(n-1)x}{n-1} \right)_{\pi/2}^{\pi} \right\} \\
 &= \frac{1}{\pi} \left\{ \frac{1}{n+1} \left(\sin \frac{(n+1)\pi}{2} - 0 \right) + \frac{1}{n-1} \left(\sin \frac{(n-1)\pi}{2} - 0 \right) \right. \\
 &\quad \left. - \frac{1}{n+1} \left(0 - \sin \frac{(n+1)\pi}{2} \right) - \frac{1}{n-1} \left(0 - \sin \frac{(n-1)\pi}{2} \right) \right\} \\
 &= \frac{1}{\pi} \left\{ \frac{2}{n+1} \sin \frac{(n+1)\pi}{2} + \frac{2}{n-1} \sin \frac{(n-1)\pi}{2} \right\} \\
 &= \frac{2}{\pi} \left[\frac{1}{n+1} \left(\sin \frac{n\pi}{2} \cos \frac{\pi}{2} + \cos \frac{n\pi}{2} \sin \frac{\pi}{2} \right) + \frac{1}{n-1} \left(\sin \frac{n\pi}{2} \cos \frac{\pi}{2} - \cos \frac{n\pi}{2} \sin \frac{\pi}{2} \right) \right]
 \end{aligned}$$

Since $\cos \frac{\pi}{2} = 0$ and $\sin \frac{\pi}{2} = 1$,

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \left[\frac{1}{n+1} \cos \frac{n\pi}{2} - \frac{1}{n-1} \cos \frac{n\pi}{2} \right] = \frac{2}{\pi} \cos \frac{n\pi}{2} \left[\frac{1}{n+1} - \frac{1}{n-1} \right] \\
 &= \frac{2}{\pi} \cos \frac{n\pi}{2} \left[\frac{(n-1) - (n+1)}{(n+1)(n-1)} \right] = \frac{2}{\pi} \cos \frac{n\pi}{2} \left(\frac{-2}{n^2-1} \right)
 \end{aligned}$$

$$a_n = \frac{-4}{\pi(n^2 - 1)} \cos \frac{n\pi}{2}, \quad \text{provided } n \neq 1$$

Evaluation of a_1 (the case $n = 1$)

$$\begin{aligned} a_1 &= \frac{2}{\pi} \int_0^\pi f(x) \cos x \, dx \\ &= \frac{2}{\pi} \left[\int_0^{\pi/2} \cos x \cos x \, dx - \int_{\pi/2}^\pi \cos x \cos x \, dx \right] \\ &= \frac{2}{\pi} \left\{ \int_0^{\pi/2} \cos^2 x \, dx - \int_{\pi/2}^\pi \cos^2 x \, dx \right\} \\ &= \frac{2}{\pi} \left\{ \int_0^{\pi/2} \left(\frac{1 + \cos 2x}{2} \right) dx - \int_{\pi/2}^\pi \left(\frac{1 + \cos 2x}{2} \right) dx \right\} \\ &= \frac{2}{\pi} \left\{ \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_0^{\pi/2} - \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_{\pi/2}^\pi \right\} \\ &= \frac{2}{\pi} \left\{ \frac{1}{2} \left[\frac{\pi}{2} + \frac{\sin \pi}{2} - 0 \right] - \frac{1}{2} \left[\left(\pi + \frac{\sin 2\pi}{2} \right) - \left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) \right] \right\} \\ &= \frac{2}{\pi} \left\{ \frac{1}{2} \cdot \frac{\pi}{2} - \frac{1}{2} \left(\pi - \frac{\pi}{2} \right) \right\} = \frac{2}{\pi} \left[\frac{\pi}{4} - \frac{\pi}{4} \right] \end{aligned}$$

$$a_1 = 0$$

Fourier series

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx = \frac{a_0}{2} + a_1 \cos x + \sum_{n=2}^{\infty} a_n \cos nx \\ &= \frac{4/\pi}{2} + 0 + \sum_{n=2}^{\infty} \frac{-4}{\pi(n^2 - 1)} \cos \frac{n\pi}{2} \cos nx \end{aligned}$$

Required Fourier series

$$|\cos x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=2}^{\infty} \frac{\cos \frac{n\pi}{2}}{n^2 - 1} \cos nx$$

4 Change of Length of the Interval

Fourier series over an interval of length 2ℓ

If $f(x)$ is defined in the interval $(\alpha, \alpha + 2\ell)$ instead of $(\alpha, \alpha + 2\pi)$, then the Fourier series

expansion of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell},$$

where

$$\begin{aligned} a_0 &= \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) dx, \\ a_n &= \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) \cos \frac{n\pi x}{\ell} dx, \\ b_n &= \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) \sin \frac{n\pi x}{\ell} dx. \end{aligned}$$

Note

- If we put $\alpha = 0$ and $\alpha = -\ell$ we get the Fourier series expansion of $f(x)$ in the intervals $(0, 2\ell)$ and $(-\ell, \ell)$ respectively, of periodicity 2ℓ .
- Similarly we can obtain the even- and odd-function Fourier series expansions in the interval $(-\ell, \ell)$:

$$\text{even: } a_0 = \frac{2}{\ell} \int_0^{\ell} f(x) dx, \quad a_n = \frac{2}{\ell} \int_0^{\ell} f(x) \cos \frac{n\pi x}{\ell} dx, \quad b_n = 0;$$

$$\text{odd: } b_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin \frac{n\pi x}{\ell} dx, \quad a_0 = a_n = 0.$$

4.1 Problem 1 : $f(x) = 3x - x^2$, $0 < x < 3$, of periodicity 3

Problem 1

Find the Fourier series expansion of $f(x) = 3x - x^2$, $0 < x < 3$, of periodicity 3.

Solution. Given $f(x) = 3x - x^2$, $0 < x < 3$, of periodicity 3.

$$2\ell = 3 \quad \Rightarrow \quad \boxed{\ell = \frac{3}{2}}$$

$\therefore$ The Fourier series is

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{3/2} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{3/2} \\ \therefore f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2n\pi x}{3} + \sum_{n=1}^{\infty} b_n \sin \frac{2n\pi x}{3} \end{aligned} \quad (1)$$

Evaluation of a_0

$$\begin{aligned}
 a_0 &= \frac{1}{\ell} \int_0^{2\ell} f(x) dx = \frac{1}{3/2} \int_0^3 (3x - x^2) dx \\
 &= \frac{2}{3} \left[3 \left(\frac{x^2}{2} \right)_0^3 - \left(\frac{x^3}{3} \right)_0^3 \right] \\
 &= \frac{2}{3} \left[\frac{3}{2}(9 - 0) - \frac{1}{3}(27 - 0) \right] \\
 &= \frac{2}{3} \left[\frac{27}{2} - \frac{27}{3} \right] = \frac{2}{3} \left[\frac{81 - 54}{6} \right] = \frac{2}{3} \cdot \frac{27}{6} = \frac{27}{9}
 \end{aligned}$$

$$a_0 = 3$$

Evaluation of a_n

$$\begin{aligned}
 a_n &= \frac{1}{\ell} \int_0^{2\ell} f(x) \cos \frac{n\pi x}{\ell} dx = \frac{1}{3/2} \int_0^3 (3x - x^2) \cos \frac{n\pi x}{3/2} dx \quad (\because 2\ell = 3) \\
 &= \frac{2}{3} \int_0^3 (3x - x^2) \cos \frac{2n\pi x}{3} dx
 \end{aligned}$$

Applying Bernoulli's rule with $u = 3x - x^2$ (so $u' = 3 - 2x$, $u'' = -2$) and $v = \cos \frac{2n\pi x}{3}$:

$$\begin{aligned}
 a_n &= \frac{2}{3} \left[(3x - x^2) \left(\frac{\sin \frac{2n\pi x}{3}}{\frac{2n\pi}{3}} \right) - (3 - 2x) \left(\frac{-\cos \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^2} \right) + (-2) \left(\frac{-\sin \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^3} \right) - 0 \right]_0^3 \\
 &= \frac{2}{3} \left[\frac{3}{2n\pi} \left((3x - x^2) \sin \frac{2n\pi x}{3} \right)_0^3 + \frac{9}{4n^2\pi^2} \left((3 - 2x) \cos \frac{2n\pi x}{3} \right)_0^3 \right. \\
 &\quad \left. + \frac{54}{8n^3\pi^3} \left(\sin \frac{2n\pi x}{3} \right)_0^3 \right]
 \end{aligned}$$

At $x = 3$: $\sin 2n\pi = 0$, $\cos 2n\pi = 1$; at $x = 0$: $\sin 0 = 0$, $\cos 0 = 1$. Also $3x - x^2 = 0$ at both $x = 0$ and $x = 3$. Hence

$$\begin{aligned}
 a_n &= \frac{2}{3} \left[0 + \frac{9}{4n^2\pi^2} \left((3 - 6)(1) - 3(1) \right) + 0 \right] \\
 &= \frac{2}{3} \left(\frac{9(-6)}{4n^2\pi^2} \right) = \frac{2}{3} \left(\frac{-54}{4n^2\pi^2} \right) = \frac{2}{3} \left(\frac{-27}{2n^2\pi^2} \right)
 \end{aligned}$$

$$a_n = \frac{-9}{n^2\pi^2}$$

Evaluation of b_n

$$\begin{aligned}
 b_n &= \frac{1}{\ell} \int_0^{2\ell} f(x) \sin \frac{n\pi x}{\ell} dx = \frac{1}{3/2} \int_0^3 (3x - x^2) \frac{\sin n\pi x}{3/2} dx \quad (\because 2\ell = 3) \\
 &= \frac{2}{3} \int_0^3 (3x - x^2) \sin \frac{2n\pi x}{3} dx \\
 &= \frac{2}{3} \left[(3x - x^2) \left(\frac{-\cos \frac{2n\pi x}{3}}{\frac{2n\pi}{3}} \right) - (3 - 2x) \left(\frac{-\sin \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3}\right)^2} \right) + (-2) \left(\frac{\cos \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3}\right)^3} \right) - 0 \right]_0^3 \\
 &= \frac{2}{3} \left[\frac{-3}{2n\pi} \left((3x - x^2) \cos \frac{2n\pi x}{3} \right)_0^3 + \frac{9}{4n^2\pi^2} \left((3 - 2x) \sin \frac{2n\pi x}{3} \right)_0^3 \right. \\
 &\quad \left. - \frac{54}{8n^3\pi^3} \left(\cos \frac{2n\pi x}{3} \right)_0^3 \right] \\
 &= \frac{2}{3} \left[\frac{-3}{2n\pi} (0 - 0) + 0 - \frac{54}{8n^3\pi^3} (1 - 1) \right]
 \end{aligned}$$

$$b_n = 0$$

Fourier series

By (1),

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2n\pi x}{3} + \sum_{n=1}^{\infty} b_n \sin \frac{2n\pi x}{3} \\
 &= \frac{3}{2} + \sum_{n=1}^{\infty} \frac{-9}{n^2\pi^2} \cos \frac{2n\pi x}{3} + 0
 \end{aligned}$$

Required Fourier series

$$f(x) = 1.5 - \frac{9}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{2n\pi x}{3}$$

4.2 Problem 2 : $f(x) = \ell - x$ and $\ell + x$ on $(-\ell, \ell)$ **Problem 2**

Find the Fourier series expansion of

$$f(x) = \begin{cases} \ell - x, & -\ell \leq x \leq 0 \\ \ell + x, & 0 \leq x \leq \ell \end{cases}$$

Solution. Given $f(x) = \begin{cases} \ell - x, & -\ell \leq x \leq 0 \\ \ell + x, & 0 \leq x \leq \ell \end{cases}$

To check even or odd

$$\begin{aligned} f(-x) &= \begin{cases} \ell + x, & -\ell \leq -x \leq 0 \\ \ell - x, & 0 \leq -x \leq \ell \end{cases} = \begin{cases} \ell + x, & \ell \geq x \geq 0 \\ \ell - x, & 0 \geq x \geq -\ell \end{cases} \\ &= \begin{cases} \ell + x, & 0 \leq x \leq \ell \\ \ell - x, & -\ell \leq x \leq 0 \end{cases} = f(x) \end{aligned}$$

$\therefore f(-x) = f(x)$: the given $f(x)$ is an **even function**.

$\therefore$ Its Fourier series is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell}$.

Evaluation of a_0

$$\begin{aligned} a_0 &= \frac{2}{\ell} \int_0^{\ell} f(x) dx = \frac{2}{\ell} \int_0^{\ell} (\ell + x) dx \\ &= \frac{2}{\ell} \left(\ell(x)_0^{\ell} + \left(\frac{x^2}{2} \right)_0^{\ell} \right) = \frac{2}{\ell} \left(\ell^2 + \frac{\ell^2}{2} \right) = \frac{2}{\ell} \left(\frac{3\ell^2}{2} \right) \end{aligned}$$

$$a_0 = 3\ell$$

Evaluation of a_n

$$\begin{aligned} a_n &= \frac{2}{\ell} \int_0^{\ell} f(x) \cos \frac{n\pi x}{\ell} dx = \frac{2}{\ell} \int_0^{\ell} (\ell + x) \cos \frac{n\pi x}{\ell} dx \\ &= \frac{2}{\ell} \left[(\ell + x) \left(\frac{\sin \frac{n\pi x}{\ell}}{\frac{n\pi}{\ell}} \right) - 1 \cdot \left(\frac{-\cos \frac{n\pi x}{\ell}}{\frac{n^2\pi^2}{\ell^2}} \right) + 0 \right]_0^{\ell} \\ &= \frac{2}{\ell} \left[\frac{\ell}{n\pi} \left((\ell + x) \sin \frac{n\pi x}{\ell} \right)_0^{\ell} + \frac{\ell^2}{n^2\pi^2} \left(\cos \frac{n\pi x}{\ell} \right)_0^{\ell} \right] \\ &= \frac{2}{\ell} \left[\frac{\ell}{n\pi} (0 - 0) + \frac{\ell^2}{n^2\pi^2} ((-1)^n - 1) \right] \quad (\because \cos n\pi = (-1)^n) \\ &= \frac{2}{\ell} \cdot \frac{\ell^2}{n^2\pi^2} ((-1)^n - 1) = \frac{2\ell}{n^2\pi^2} ((-1)^n - 1) \end{aligned}$$

$$a_n = \begin{cases} \frac{-4\ell}{n^2\pi^2}, & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases}$$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell} = \frac{3\ell}{2} + \sum_{n=1,3,5,\dots}^{\infty} \frac{-4\ell}{n^2\pi^2} \cos \frac{n\pi x}{\ell}$$

Required Fourier series

$$f(x) = \frac{3\ell}{2} - \frac{4\ell}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{\ell}$$

5 Half Range Fourier Series**Half range series**

So far we have expressed Fourier series for a function $f(x)$ defined over a length 2π or 2ℓ . Suppose instead $f(x)$ is defined over only the **half length** π or ℓ , i.e. in the interval $(0, \pi)$ or $(0, \ell)$. Then its Fourier series contains either *fully cosine terms* or *fully sine terms*.

- If we **extend** the given function as an **even function** in the interval $(-\pi, \pi)$ or $(-\ell, \ell)$, we get a **half range cosine series**.
- If we **extend** the given function as an **odd function** in the interval $(-\pi, \pi)$ or $(-\ell, \ell)$, we get a **half range sine series**.

Half range formulae in $(0, \ell)$

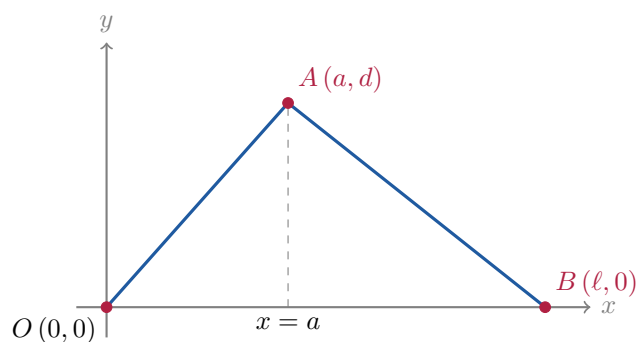
Cosine series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell}, \quad a_0 = \frac{2}{\ell} \int_0^{\ell} f(x) dx, \quad a_n = \frac{2}{\ell} \int_0^{\ell} f(x) \cos \frac{n\pi x}{\ell} dx$

Sine series: $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell}, \quad b_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin \frac{n\pi x}{\ell} dx$

5.1 Problem 1 : Function defined by the graph OAB **Problem 1**

For the function defined by the graph OAB , show that its half range Fourier sine series is

$$f(x) = \frac{2d\ell^2}{a\pi^2(\ell - a)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi a}{\ell} \sin \frac{n\pi x}{\ell}.$$



Solution. From the given graph OAB we have

$$f(x) = \begin{cases} \text{Equation of straight line } OA, & 0 \leq x \leq a \\ \text{Equation of straight line } AB, & a \leq x \leq \ell \end{cases} \quad (1)$$

Equation of the straight line OA

Through $O(0, 0)$ and $A(a, d)$, using $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$:

$$\frac{y - 0}{d - 0} = \frac{x - 0}{a - 0}$$

$$\frac{y}{d} = \frac{x}{a}$$

$$y = \frac{d}{a}x \quad (\text{line } OA)$$

Equation of the straight line AB

Through $A(a, d)$ and $B(\ell, 0)$:

$$\frac{y - d}{0 - d} = \frac{x - a}{\ell - a}$$

$$\frac{y - d}{-d} = \frac{x - a}{\ell - a}$$

$$y - d = \frac{d}{a - \ell}(x - a)$$

$$y = d + \frac{dx}{a - \ell} - \frac{ad}{a - \ell} = \frac{ad - \ell d + dx - ad}{a - \ell} = \frac{d(x - \ell)}{a - \ell}$$

$$y = \frac{d}{a - \ell}(x - \ell) \quad (\text{line } AB)$$

$\therefore$ From (1),

$$f(x) = \begin{cases} \frac{d}{a}x, & 0 \leq x \leq a \\ \frac{d}{a - \ell}(x - \ell), & a \leq x \leq \ell \end{cases}$$

Half range Fourier sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell}, \quad b_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin \frac{n\pi x}{\ell} dx$$

$$\begin{aligned}
b_n &= \frac{2}{\ell} \left\{ \int_0^a f(x) \sin \frac{n\pi x}{\ell} dx + \int_a^\ell f(x) \sin \frac{n\pi x}{\ell} dx \right\} \\
&= \frac{2}{\ell} \left\{ \int_0^a \frac{d}{a} x \sin \frac{n\pi x}{\ell} dx + \int_a^\ell \frac{d}{a-\ell} (x-\ell) \sin \frac{n\pi x}{\ell} dx \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d}{a} \int_0^a \underbrace{x}_u \underbrace{\sin \frac{n\pi x}{\ell}}_v dx + \frac{d}{a-\ell} \int_a^\ell \underbrace{(x-\ell)}_u \underbrace{\sin \frac{n\pi x}{\ell}}_v dx \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d}{a} \left[x \left(\frac{-\cos \frac{n\pi x}{\ell}}{\frac{n\pi}{\ell}} \right) - 1 \cdot \left(\frac{-\sin \frac{n\pi x}{\ell}}{\frac{n^2\pi^2}{\ell^2}} \right) + 0 \right]_0^a \right. \\
&\quad \left. + \frac{d}{a-\ell} \left[(x-\ell) \left(\frac{-\cos \frac{n\pi x}{\ell}}{\frac{n\pi}{\ell}} \right) - (1) \left(\frac{-\sin \frac{n\pi x}{\ell}}{\frac{n^2\pi^2}{\ell^2}} \right) + 0 \right]_a^\ell \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d}{a} \left[-\frac{\ell}{n\pi} \left(x \cos \frac{n\pi x}{\ell} \right)_0^a + \frac{\ell^2}{n^2\pi^2} \left(\sin \frac{n\pi x}{\ell} \right)_0^a \right] \right. \\
&\quad \left. + \frac{d}{a-\ell} \left[-\frac{\ell}{n\pi} \left((x-\ell) \cos \frac{n\pi x}{\ell} \right)_a^\ell + \frac{\ell^2}{n^2\pi^2} \left(\sin \frac{n\pi x}{\ell} \right)_a^\ell \right] \right\}
\end{aligned}$$

Applying the limits (note $\sin n\pi = 0$, and $(x-\ell) = 0$ at $x = \ell$):

$$\begin{aligned}
b_n &= \frac{2}{\ell} \left\{ \frac{d}{a} \left[-\frac{\ell}{n\pi} \left(a \cos \frac{n\pi a}{\ell} - 0 \right) + \frac{\ell^2}{n^2\pi^2} \left(\sin \frac{n\pi a}{\ell} - 0 \right) \right] \right. \\
&\quad \left. + \frac{d}{a-\ell} \left[-\frac{\ell}{n\pi} \left(0 - (a-\ell) \cos \frac{n\pi a}{\ell} \right) + \frac{\ell^2}{n^2\pi^2} \left(0 - \sin \frac{n\pi a}{\ell} \right) \right] \right\} \\
&= \frac{2}{\ell} \left\{ -\frac{d\ell}{n\pi} \cos \frac{n\pi a}{\ell} + \frac{d\ell^2}{an^2\pi^2} \sin \frac{n\pi a}{\ell} + \frac{d\ell}{n\pi} \cos \frac{n\pi a}{\ell} - \frac{d}{a-\ell} \cdot \frac{\ell^2}{n^2\pi^2} \sin \frac{n\pi a}{\ell} \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d\ell^2}{n^2\pi^2} \sin \frac{n\pi a}{\ell} \left(\frac{1}{a} - \frac{1}{a-\ell} \right) \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d\ell^2}{n^2\pi^2} \sin \frac{n\pi a}{\ell} \left(\frac{a-\ell-a}{a(a-\ell)} \right) \right\} \\
&= \frac{2}{\ell} \left\{ \frac{d\ell^2}{n^2\pi^2} \sin \frac{n\pi a}{\ell} \left(\frac{\ell}{a(\ell-a)} \right) \right\}
\end{aligned}$$

$$b_n = \frac{2}{a(\ell-a)} \cdot \frac{d\ell^2}{n^2\pi^2} \sin \frac{n\pi a}{\ell}$$

Fourier series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell} = \sum_{n=1}^{\infty} \frac{2d\ell^2}{a(\ell-a)n^2\pi^2} \sin \frac{n\pi a}{\ell} \sin \frac{n\pi x}{\ell}$$

Required result

$$f(x) = \frac{2d\ell^2}{\pi^2 a(\ell - a)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi a}{\ell} \sin \frac{n\pi x}{\ell}$$

5.2 Problem 2 : Half range cosine series of $f(x) = x$ in $(0, \pi)$ **Problem 2**

Find the half range cosine series for the function $f(x) = x$ in $(0, \pi)$.

Solution. The half range cosine series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx.$$

Evaluation of a_0

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{\pi^2}{2} \right]$$

$$a_0 = \pi$$

Evaluation of a_n

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx \\ &= \frac{2}{\pi} \left[x \frac{\sin nx}{n} + (1) \frac{\cos nx}{n^2} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\frac{x}{n} \sin nx + \frac{\cos nx}{n^2} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\left(\frac{\pi}{n} \sin n\pi - 0 \right) + \left(\frac{\cos n\pi}{n^2} - \frac{\cos 0}{n^2} \right) \right] \\ &= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] \quad (\because \sin n\pi = 0, \cos n\pi = (-1)^n) \end{aligned}$$

$$a_n = \frac{2}{n^2\pi} [(-1)^n - 1] = \begin{cases} 0, & n \text{ even} \\ -\frac{4}{n^2\pi}, & n \text{ odd} \end{cases}$$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} [(-1)^n - 1] \cos nx$$

Required half range cosine series

$$x = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\cos nx}{n^2} = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

6 Parseval's Identity and Root Mean Square Value**Parseval's Identity**

If $f(x)$ is defined in the interval $(0, 2\ell)$, Parseval's identity is

$$\frac{1}{2\ell} \int_0^{2\ell} (f(x))^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Similarly, in the interval $(-\ell, \ell)$ Parseval's identity is given by

$$\frac{1}{2\ell} \int_{-\ell}^{\ell} (f(x))^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Special cases in $(-\ell, \ell)$

If $f(x)$ is an **even function** in $(-\ell, \ell)$, its Parseval's identity is

$$\frac{1}{\ell} \int_0^{\ell} (f(x))^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2.$$

If $f(x)$ is an **odd function** in $(-\ell, \ell)$, its Parseval's identity is

$$\frac{1}{\ell} \int_0^{\ell} (f(x))^2 dx = \frac{1}{2} \sum_{n=1}^{\infty} b_n^2.$$

Note

In the above results, if we put $\ell = \pi$ we get Parseval's identity in the intervals $(0, 2\pi)$ and $(-\pi, \pi)$.

6.1 Root Mean Square Value (RMS Value)**RMS value**

For a function $y = f(x)$ defined in the interval (a, b) , its RMS value is defined by

$$\bar{y}^2 = \frac{1}{b-a} \int_a^b (f(x))^2 dx.$$

By comparing with Parseval's identity we see that

$$\bar{y}^2 = \text{Parseval's identity in the respective interval.}$$

6.2 Problem 1 : $f(x) = x^2$ in $(-\pi, \pi)$ – deduce $\pi^4/90$ **Problem 1**

Find the Fourier series for $f(x) = x^2$ in the interval $-\pi \leq x \leq \pi$ and hence deduce that

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^4}{90}.$$

Solution. Given $f(x) = x^2$, $-\pi \leq x \leq \pi$.

Already we know (from Section 2, Problem 4) that

$$a_0 = \frac{2}{3}\pi^2, \quad a_n = \frac{4}{n^2}(-1)^n, \quad b_n = 0,$$

and also that $f(x)$ is an **even function**.

$\therefore$ By Parseval's identity for an even function $f(x) = x^2$ in $(-\pi, \pi)$:

$$\begin{aligned} \frac{1}{\pi} \int_0^\pi (f(x))^2 dx &= \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 \\ \Rightarrow \frac{1}{\pi} \int_0^\pi (x^2)^2 dx &= \frac{1}{4} \cdot \frac{4}{9}\pi^4 + \frac{1}{2} \sum_{n=1}^{\infty} \frac{16}{n^4}(-1)^{2n} \\ \frac{1}{\pi} \int_0^\pi x^4 dx &= \frac{\pi^4}{9} + 8 \sum_{n=1}^{\infty} \frac{1}{n^4} \quad (\because (-1)^{2n} = 1) \\ \frac{1}{\pi} \left(\frac{x^5}{5} \right)_0^\pi &= \frac{\pi^4}{9} + 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots \right] \\ \frac{1}{\pi} \left(\frac{\pi^5}{5} \right) &= \frac{\pi^4}{9} + 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots \right] \\ \frac{\pi^4}{5} - \frac{\pi^4}{9} &= 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots \right] \\ \frac{9\pi^4 - 5\pi^4}{45} &= \frac{4\pi^4}{45} = 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots \right] \\ \therefore \quad \boxed{\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots} \end{aligned}$$

Note

The RMS value is generally useful to identify the **effective voltage** or **effective current** in an electrical circuit that carries alternating current.

6.3 Problem 2 : Effective voltage of an alternating current

Problem 2

The instantaneous voltage e of an alternating current is given by

$$e = E_1 \sin(\omega t + \beta_1) + E_3 \sin(3\omega t + \beta_3) + E_5 \sin(5\omega t + \beta_5) + \dots$$

Show that the effective voltage $\bar{e}$ is given by $\frac{1}{2}(E_1^2 + E_3^2 + E_5^2 + \dots) = \bar{e}^2$.

Solution. Given the instantaneous voltage of the AC current wave

$$e = E_1 \sin(\omega t + \beta_1) + E_3 \sin(3\omega t + \beta_3) + E_5 \sin(5\omega t + \beta_5) + \dots$$

Expanding each term using $\sin(A + B) = \sin A \cos B + \cos A \sin B$:

$$\begin{aligned} e &= E_1 [\sin \omega t \cos \beta_1 + \cos \omega t \sin \beta_1] + E_3 [\sin 3\omega t \cos \beta_3 + \cos 3\omega t \sin \beta_3] \\ &\quad + E_5 [\sin 5\omega t \cos \beta_5 + \cos 5\omega t \sin \beta_5] + \dots \end{aligned}$$

Grouping the sine and cosine terms:

$$\begin{aligned} e &= (E_1 \cos \beta_1) \sin \omega t + (E_3 \cos \beta_3) \sin 3\omega t + (E_5 \cos \beta_5) \sin 5\omega t + \dots \\ &\quad + (E_1 \sin \beta_1) \cos \omega t + (E_3 \sin \beta_3) \cos 3\omega t + (E_5 \sin \beta_5) \cos 5\omega t + \dots \\ &= \sum_{n=1,3,5,\dots} (E_n \cos \beta_n) \sin n\omega t + \sum_{n=1,3,5,\dots} (E_n \sin \beta_n) \cos n\omega t \end{aligned}$$

Comparing with the Fourier form

$$e = \sum_{n=1,3,5,\dots} b_n \sin n\omega t + \sum_{n=1,3,5,\dots} a_n \cos n\omega t,$$

we get

$$a_n = E_n \sin \beta_n \quad \text{and} \quad b_n = E_n \cos \beta_n, \quad a_0 = 0.$$

$\therefore$ By the RMS value, the effective voltage is

$$\bar{e}^2 = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad (\because \text{RMS value} = \text{Parseval's identity})$$

$$\bar{e}^2 = 0 + \frac{1}{2} \sum_{n=1,3,5,\dots} [(E_n \sin \beta_n)^2 + (E_n \cos \beta_n)^2]$$

$$= \frac{1}{2} \sum_{n=1,3,5,\dots} E_n^2 (\sin^2 \beta_n + \cos^2 \beta_n)$$

$$= \frac{1}{2} \sum_{n=1,3,5,\dots} E_n^2 \quad (\because \sin^2 \theta + \cos^2 \theta = 1)$$

$$\therefore \boxed{\bar{e}^2 = \frac{1}{2} [E_1^2 + E_3^2 + E_5^2 + \dots]}$$

7 Complex (Exponential) Form of Fourier Series

Complex or exponential form

If $f(x)$ is defined in $(\alpha, \alpha + 2\ell)$, then its complex or exponential form of Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/\ell}, \quad C_n = \frac{1}{2\ell} \int_{\alpha}^{\alpha+2\ell} f(x) e^{-in\pi x/\ell} dx.$$

Deductions from the definition

In $(0, 2\ell)$:

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/\ell}, \quad C_n = \frac{1}{2\ell} \int_0^{2\ell} f(x) e^{-in\pi x/\ell} dx$$

Similarly in $(-\ell, \ell)$:

$$C_n = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-in\pi x/\ell} dx$$

In $(0, 2\pi)$:

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}, \quad C_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx$$

Similarly in $(-\pi, \pi)$:

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

7.1 Example 1 : $f(x) = \cos(ax)$ in $(-\pi, \pi)$ **Example 1**

Discuss the complex form of Fourier series for $f(x) = \cos(ax)$ in $(-\pi, \pi)$ for

1. a is an integer,
2. a is not an integer.

Solution. Given $f(x) = \cos(ax)$ in $(-\pi, \pi)$. Its complex form of Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}, \quad C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

With $f(x) = \cos(ax)$,

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(ax) e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \cos(ax) dx$$

Using the standard result $\int e^{px} \cos qx \, dx = \frac{e^{px}}{p^2 + q^2} (p \cos qx + q \sin qx)$ with $p = -in$ and $q = a$:

$$\begin{aligned}
 C_n &= \frac{1}{2\pi} \left[\frac{e^{-inx}}{(-in)^2 + a^2} \left((-in) \cos ax + a \sin ax \right) \right]_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi} \left[\frac{e^{-inx}}{a^2 - n^2} \left(-in \cos ax + a \sin ax \right) \right]_{-\pi}^{\pi} \quad (\because (-in)^2 = -n^2) \\
 &= \frac{1}{2\pi} \left[\frac{e^{-in\pi}}{a^2 - n^2} \left(-in \cos(a\pi) + a \sin(a\pi) \right) - \frac{e^{in\pi}}{a^2 - n^2} \left(-in \cos(-a\pi) + a \sin(-a\pi) \right) \right] \\
 &= \frac{1}{2\pi(a^2 - n^2)} \left[e^{-in\pi} \left(-in \cos(a\pi) + a \sin(a\pi) \right) - e^{in\pi} \left(-in \cos(a\pi) - a \sin(a\pi) \right) \right]
 \end{aligned}$$

Case (i) : a is an integer

$$C_n = \frac{1}{2\pi} \left[\frac{e^{-in\pi}}{a^2 - n^2} \left(-in \cos(a\pi) + a \sin(a\pi) \right) - \frac{e^{in\pi}}{a^2 - n^2} \left(-in \cos(a\pi) - a \sin(a\pi) \right) \right]$$

Since a is an integer, $\sin(a\pi) = 0$, and therefore

$$\begin{aligned}
 C_n &= \frac{1}{2\pi} \cdot \frac{-in \cos(a\pi)}{a^2 - n^2} (e^{-in\pi} - e^{in\pi}) \\
 &= \frac{-in \cos(a\pi)}{2\pi(a^2 - n^2)} (-2i \sin(n\pi)) \quad (\because e^{-i\theta} - e^{i\theta} = -2i \sin \theta) \\
 &= \frac{-n \cos(a\pi) \sin(n\pi)}{\pi(a^2 - n^2)}
 \end{aligned}$$

Since $\sin(n\pi) = 0$ for integer n ,

$$C_n = 0$$

Thus there is **no possibility of a complex Fourier series** of $f(x) = \cos(ax)$ when a is an integer.

Case (ii) : a is not an integer

$$\begin{aligned}
 C_n &= \frac{1}{2\pi(a^2 - n^2)} \left[-in \cos(a\pi) (e^{-in\pi} - e^{in\pi}) + a \sin(a\pi) (e^{-in\pi} + e^{in\pi}) \right] \\
 &= \frac{1}{2\pi(a^2 - n^2)} \left[-in \cos(a\pi) (-2i \sin(n\pi)) + a \sin(a\pi) (2 \cos(n\pi)) \right] \\
 &= \frac{1}{2\pi(a^2 - n^2)} \left[-2n \cos(a\pi) \sin(n\pi) + 2a \sin(a\pi) \cos(n\pi) \right] \\
 &= \frac{1}{\pi(a^2 - n^2)} \left[0 + a(-1)^n \sin(a\pi) \right] \quad (\because \sin(n\pi) = 0, \cos(n\pi) = (-1)^n)
 \end{aligned}$$

$$C_n = \frac{(-1)^n a \sin(a\pi)}{\pi(a^2 - n^2)}$$

Therefore the complex form of Fourier series is $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$, and for a not an integer:

Required complex Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{a(-1)^n \sin(a\pi)}{\pi(a^2 - n^2)} e^{inx}, \quad \text{where } a \text{ is not an integer.}$$

7.2 Example 2 : $f(x) = e^{ax}$ in $(-\ell, \ell)$ **Example 2**

If $f(x) = e^{ax}$ in $(-\ell, \ell)$, then $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/\ell}$, where a is not an integer and $C_n = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-in\pi x/\ell} dx$. Find C_n .

Solution. Given $f(x) = e^{ax}$.

$$\begin{aligned} C_n &= \frac{1}{2\ell} \int_{-\ell}^{\ell} e^{ax} e^{-in\pi x/\ell} dx = \frac{1}{2\ell} \int_{-\ell}^{\ell} e^{(a - \frac{in\pi}{\ell})x} dx \\ &= \frac{1}{2\ell} \left[\frac{e^{(a - \frac{in\pi}{\ell})x}}{a - \frac{in\pi}{\ell}} \right]_{-\ell}^{\ell} \\ &= \frac{1}{2\ell} \cdot \frac{1}{a - \frac{in\pi}{\ell}} \left[e^{(a - \frac{in\pi}{\ell})\ell} - e^{(a - \frac{in\pi}{\ell})(-\ell)} \right] \\ &= \frac{1}{2\ell (a - \frac{in\pi}{\ell})} \left[e^{a\ell - in\pi} - e^{-a\ell + in\pi} \right] \end{aligned}$$

Noting

$e^{-in\pi} = (-1)^n$ and $e^{in\pi} = (-1)^n$, so we get

$$\begin{aligned} e^{a\ell - in\pi} - e^{-a\ell + in\pi} &= (-1)^n (e^{a\ell} - e^{-a\ell}) \\ &= 2(-1)^n \sinh(a\ell) \quad \left(\because \sinh \theta = \frac{e^{\theta} - e^{-\theta}}{2} \right) \end{aligned}$$

Hence

$$C_n = \frac{2(-1)^n \sinh(a\ell)}{2\ell (a - \frac{in\pi}{\ell})} = \frac{(-1)^n \sinh(a\ell)}{\ell (a - \frac{in\pi}{\ell})}$$

Simplified form

$$C_n = \frac{(-1)^n \sinh(a\ell)}{a\ell - in\pi}$$

Therefore the complex form of Fourier series is $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/\ell}$.

7.3 Problem : $f(x) = 0$ on $(-\pi, 0)$ and $\sin x$ on $(0, \pi)$ **Problem**

Find the Fourier series for the function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \sin x, & 0 < x < \pi \end{cases}$$

and hence evaluate $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots = \frac{\pi - 2}{4}$.**Solution.** Given $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \sin x, & 0 < x < \pi \end{cases}$

Its Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Evaluation of a_0

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} \sin x dx \right] \\ &= \frac{1}{\pi} [-\cos x]_0^{\pi} = \frac{1}{\pi} [-\cos \pi + \cos 0] \\ &= \frac{1}{\pi} [-(-1) + 1] \end{aligned}$$

$$a_0 = \frac{2}{\pi}$$

Evaluation of a_n (for $n \neq 1$)

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 0 \cdot \cos nx dx + \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx dx \\ &= \frac{1}{\pi} \int_0^{\pi} \underbrace{\sin x}_A \underbrace{\cos nx}_B dx \end{aligned}$$

Using $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$:

$$\begin{aligned}
 a_n &= \frac{1}{2\pi} \int_0^\pi [\sin(1+n)x + \sin(1-n)x] dx \\
 &= \frac{1}{2\pi} \left[\frac{-\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n} \right]_0^\pi \\
 &= \frac{1}{2\pi} \left[\left(\frac{-\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n} \right) - \left(\frac{-1}{1+n} - \frac{1}{1-n} \right) \right] \\
 &= \frac{1}{2\pi} \left[\left(\frac{-(-1)^{n+1}}{1+n} - \frac{(-1)^{1-n}}{1-n} \right) + \left(\frac{1}{1+n} + \frac{1}{1-n} \right) \right] \\
 &= \frac{1}{2\pi} \left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right] \\
 &= \frac{1}{2\pi} \left[(-1)^n \left(\frac{2}{1-n^2} \right) + \left(\frac{2}{1-n^2} \right) \right] \quad \left(\because \frac{1}{1+n} + \frac{1}{1-n} = \frac{2}{1-n^2} \right) \\
 &= \frac{1}{2\pi} \times \frac{2}{1-n^2} [(-1)^n + 1]
 \end{aligned}$$

$$a_n = \frac{1}{\pi(1-n^2)} [(-1)^n + 1], \quad n \neq 1$$

Evaluation of b_n

$$\begin{aligned}
 b_1 &= \frac{1}{\pi} \int_0^\pi \sin x \sin x dx = \frac{1}{\pi} \int_0^\pi \sin^2 x dx = \frac{1}{\pi} \int_0^\pi \left(\frac{1 - \cos 2x}{2} \right) dx \\
 &= \frac{1}{2\pi} \left[x - \frac{\sin 2x}{2} \right]_0^\pi = \frac{1}{2\pi} [\pi - 0]
 \end{aligned}$$

$$b_1 = \frac{1}{2}$$

Fourier series

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \\
 &= \frac{2/\pi}{2} + \frac{\sin x}{2} + \frac{1}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{1-n^2} \cos nx \\
 &= \frac{1}{\pi} + \frac{1}{2} \sin x + \frac{2}{\pi} \sum_{n=2,4,6,\dots} \left[\frac{1}{(1+n)(1-n)} \cos nx \right]
 \end{aligned}$$

Required Fourier series

$$f(x) = \frac{1}{\pi} + \frac{1}{2} \sin x - \frac{2}{\pi} \left[\frac{\cos 2x}{1 \cdot 3} + \frac{\cos 4x}{3 \cdot 5} + \frac{\cos 6x}{5 \cdot 7} + \dots \right] \quad (1)$$

Deduction

Put $x = \frac{\pi}{2}$ in (1). Since $f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$:

$$\begin{aligned}
 f\left(\frac{\pi}{2}\right) &= \frac{1}{\pi} + \frac{1}{2} \sin \frac{\pi}{2} - \frac{2}{\pi} \left[\frac{\cos \pi}{1 \cdot 3} + \frac{\cos 2\pi}{3 \cdot 5} + \frac{\cos 3\pi}{5 \cdot 7} + \dots \right] \\
 1 &= \frac{1}{\pi} + \frac{1}{2} - \frac{2}{\pi} \left[\frac{-1}{1 \cdot 3} + \frac{1}{3 \cdot 5} - \frac{1}{5 \cdot 7} + \dots \right] \\
 1 - \frac{1}{2} - \frac{1}{\pi} &= \frac{2}{\pi} \left[\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right] \\
 \frac{1}{2} - \frac{1}{\pi} &= \frac{2}{\pi} \left[\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right] \\
 \frac{\pi - 2}{2\pi} \cdot \frac{\pi}{2} &= \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \\
 \therefore \quad \boxed{\frac{\pi - 2}{4} &= \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots}
 \end{aligned}$$

8 Harmonic Analysis

Harmonic analysis

Finding the Fourier series of a given function $f(x)$ through its **numerical data** is known as **harmonic analysis**.

We know that if $f(x)$ is defined in the interval $(\alpha, \alpha + 2\pi)$ then its Fourier series is given by

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \\
 &= \frac{a_0}{2} + (a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \dots) + (b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \dots) \\
 f(x) &= \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + \dots
 \end{aligned}$$

Terminology

In the above expansion:

- the term $(a_1 \cos x + b_1 \sin x)$ is known as the **first or fundamental harmonic**,
- $(a_2 \cos 2x + b_2 \sin 2x)$ is known as the **second harmonic**,
- $(a_3 \cos 3x + b_3 \sin 3x)$ is known as the **third harmonic**, and so on.

Harmonic coefficients

The harmonic coefficients $a_0, a_1, b_1, \dots$ can be calculated by the following formulae:

$$\begin{aligned} a_0 &= 2(\text{Mean value of } y) & \Rightarrow & a_0 = 2 \left(\frac{\sum y}{N} \right) \\ a_n &= 2(\text{Mean value of } y \cos nx) & \Rightarrow & a_n = 2 \left(\frac{\sum y \cos nx}{N} \right) \\ b_n &= 2(\text{Mean value of } y \sin nx) & \Rightarrow & b_n = 2 \left(\frac{\sum y \sin nx}{N} \right) \end{aligned}$$

where N is the **number of observations** considered for the calculation.

Note – amplitude and phase

In the case of an alternating current wave, we can calculate the **amplitude** A_n and **phase difference** ϕ_n for each of the harmonic series. That is, the amplitude A_n based on the n^{th} harmonic is given by

$$A_n = \sqrt{a_n^2 + b_n^2}, \quad \text{and the phase difference } \phi_n = \tan^{-1} \left(\frac{b_n}{a_n} \right).$$

8.1 Problem 1 : Fourier series up to the second harmonic**Problem 1**

Find the Fourier series up to the second harmonic for the function $f(x)$ given by the following table:

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Solution. For the given $f(x)$, defined in $(0, 2\pi)$, the Fourier series up to the 2nd harmonic is given by

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x).$$

We have the following table (the value at $x = 2\pi$ repeats the value at $x = 0$ and is therefore omitted, so $N = 6$):

x	$y = f(x)$	$\cos x$	$\cos 2x$	$\sin x$	$\sin 2x$	$y \cos x$	$y \cos 2x$	$y \sin x$	$y \sin 2x$
0	1.0	1	1	0	0	1	1	0	0
$\pi/3$	1.4	0.5	-0.5	0.866	0.866	0.7	-0.7	1.2124	1.2124
$2\pi/3$	1.9	-0.5	-0.5	0.866	-0.866	-0.95	-0.95	1.6454	-1.6454
π	1.7	-1	1	0	0	-1.7	1.7	0	0
$4\pi/3$	1.5	-0.5	-0.5	-0.866	0.866	-0.75	-0.75	-1.299	1.299
$5\pi/3$	1.2	0.5	-0.5	-0.866	-0.866	0.6	-0.6	-1.0392	-1.0392
$\sum y = 8.7$						$\sum y \cos x = -1.1$	$\sum y \cos 2x = -0.3$	$\sum y \sin x = 0.519$	$\sum y \sin 2x = -0.173$

Harmonic coefficients

$$\begin{aligned}
 a_0 &= 2 \left(\frac{\sum y}{N} \right) = 2 \left(\frac{8.7}{6} \right) & \Rightarrow & \boxed{a_0 = 2.9} \\
 a_1 &= 2 \left(\frac{\sum y \cos x}{N} \right) = 2 \left(\frac{-1.1}{6} \right) & \Rightarrow & \boxed{a_1 = -0.366} \\
 a_2 &= 2 \left(\frac{\sum y \cos 2x}{N} \right) = 2 \left(\frac{-0.3}{6} \right) & \Rightarrow & \boxed{a_2 = -0.1} \\
 b_1 &= 2 \left(\frac{\sum y \sin x}{N} \right) = 2 \left(\frac{0.519}{6} \right) & \Rightarrow & \boxed{b_1 = 0.173} \\
 b_2 &= 2 \left(\frac{\sum y \sin 2x}{N} \right) = 2 \left(\frac{-0.173}{6} \right) & \Rightarrow & \boxed{b_2 = -0.0576}
 \end{aligned}$$

Fourier series

∴ The Fourier series of the second harmonic function is given by

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x)$$

Required Fourier series

$$f(x) = \frac{2.9}{2} + (-0.366 \cos x + 0.173 \sin x) + (-0.1 \cos 2x - 0.0576 \sin 2x)$$

8.2 Problem 2 : Periodic current over a period**Problem 2**

The following table gives the variations of a periodic current over a period:

t (sec)	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A (amp)	1.98	1.36	1.05	1.36	-0.88	-0.25	1.98

Show that there is a direct current path of 0.75 amplitude in the variable current. Also find the amplitude A_1 and phase difference ϕ_1 for the first harmonic.

Solution. For the given data the periodicity is 2ℓ . Therefore

$$2\ell = T \quad \Rightarrow \quad \boxed{\ell = \frac{T}{2}}$$

Its Fourier series is

$$\begin{aligned}
 f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell} \\
 &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{T/2} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{T/2} \\
 &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2n\pi x}{T} + \sum_{n=1}^{\infty} b_n \sin \frac{2n\pi x}{T}
 \end{aligned}$$

Let $\theta = \frac{2\pi x}{T}$; then

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\theta + \sum_{n=1}^{\infty} b_n \sin n\theta.$$

$\Rightarrow$ The Fourier series up to the **first harmonic** is

$$f(x) = \frac{a_0}{2} + (a_1 \cos \theta + b_1 \sin \theta), \quad \text{where } \theta = \frac{2\pi x}{T} \quad (1)$$

$$a_0 = 2 \left(\frac{\sum y}{N} \right), \quad a_1 = 2 \left(\frac{\sum y \cos \theta}{N} \right), \quad b_1 = 2 \left(\frac{\sum y \sin \theta}{N} \right)$$

We have the following table (with $N = 6$; $t = T$ repeats $t = 0$):

$t (x)$	$A (y)$	$\theta = \frac{2\pi x}{T}$	$\cos \theta$	$\sin \theta$	$y \cos \theta$	$y \sin \theta$
0	1.98	0	1	0	1.98	0
$T/6$	1.36	$\pi/3$	0.5	0.866	0.68	1.177
$T/3$	1.05	$2\pi/3$	-0.5	0.866	-0.525	0.909
$T/2$	1.36	π	-1	0	-1.36	0
$2T/3$	-0.88	$4\pi/3$	-0.5	-0.866	0.44	0.762
$5T/6$	-0.25	$5\pi/3$	0.5	-0.866	-0.125	0.216
$\sum y = 4.62$					$\sum y \cos \theta = 1.09$	$\sum y \sin \theta = 3.065$

Harmonic coefficients

$$a_0 = 2 \left(\frac{\sum y}{N} \right) = 2 \left(\frac{4.62}{6} \right) \quad \Rightarrow \quad a_0 = 1.54$$

$$a_1 = 2 \left(\frac{\sum y \cos \theta}{N} \right) = 2 \left(\frac{1.09}{6} \right) \quad \Rightarrow \quad a_1 = 0.363$$

$$b_1 = 2 \left(\frac{\sum y \sin \theta}{N} \right) = 2 \left(\frac{3.065}{6} \right) \quad \Rightarrow \quad b_1 = 1.02$$

Substituting in (1):

Fourier series up to the first harmonic

$$f(x) = 0.77 + 0.363 \cos \theta + 1.02 \sin \theta, \quad \text{where } \theta = \frac{2\pi x}{T}$$

The constant term $\frac{a_0}{2} = 0.77 \approx 0.75$ is the **direct current path** in the variable current, as required.

Amplitude and phase difference of the first harmonic**Amplitude**

$$A_1 = \sqrt{a_1^2 + b_1^2}$$

$$= \sqrt{(0.363)^2 + (1.02)^2}$$

$$A_1 = 1.083$$

Phase difference

$$\phi_1 = \tan^{-1} \left(\frac{b_1}{a_1} \right)$$

$$= \tan^{-1} \left(\frac{1.021}{0.363} \right)$$

$$\phi_1 = 1.22 \text{ radians}$$

8.3 Problem 3 : Turning moment of a steam engine**Problem 3**

A steam engine operates from the principal crank. It is observed that the turning moment $T = f(\theta)$ units on the crank shaft occurring due to the crank angle θ (in degrees) is given in the following table:

θ	0	30°	60°	90°	120°	150°	180°
T	0	5.2	8.1	7.9	5.5	2.6	0

Obtain the first three terms of the sine series to represent T .

Solution. Let $T = f(\theta)$. Its Fourier series is required up to three terms of the sine series:

$$f(\theta) = \sum b_n \sin n\theta \quad \Rightarrow \quad T = f(\theta) = b_1 \sin \theta + b_2 \sin 2\theta + b_3 \sin 3\theta$$

We have the following table (with $N = 6$; $\theta = 180^\circ$ repeats the value at $\theta = 0$):

θ	T	$\sin \theta$	$\sin 2\theta$	$\sin 3\theta$	$T \sin \theta$	$T \sin 2\theta$	$T \sin 3\theta$
0	0	0	0	0	0	0	0
30	5.2	0.5	0.866	1	2.6	4.503	5.2
60	8.1	0.866	0.866	0	7.014	7.014	0
90	7.9	1	0	-1	7.9	0	-7.9
120	5.5	0.866	-0.866	0	4.763	-4.763	0
150	2.6	0.5	-0.866	1	1.3	-2.251	2.6
180	0	0	0	0	0	0	0
					$\sum T \sin \theta$ $= 23.577$	$\sum T \sin 2\theta$ $= 4.503$	$\sum T \sin 3\theta$ $= -0.1$

Harmonic coefficients

$$b_1 = 2 \left(\frac{\sum T \sin \theta}{N} \right) = 2 \left(\frac{23.577}{6} \right) \quad \Rightarrow \quad b_1 = 7.85$$

$$b_2 = 2 \left(\frac{\sum T \sin 2\theta}{N} \right) = 2 \left(\frac{4.503}{6} \right) \quad \Rightarrow \quad b_2 = 1.50$$

$$b_3 = 2 \left(\frac{\sum T \sin 3\theta}{N} \right) = 2 \left(\frac{-0.1}{6} \right) \quad \Rightarrow \quad b_3 = -0.033$$

Required sine series

$$f(\theta) = 7.85 \sin \theta + 1.50 \sin 2\theta - 0.033 \sin 3\theta$$

8.4 Problem 4 : First three harmonics of $y = f(x)$

Problem 4

Obtain the first three harmonics of the Fourier series of $y = f(x)$ which is defined in the following table:

x	0	1	2	3	4	5	6
$y = f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Solution. For the given data the periodicity is 2ℓ :

$$2\ell = 6 \quad \Rightarrow \quad \boxed{\ell = 3}$$

Its Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{3} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{3}$$

Putting $\theta = \frac{\pi x}{3}$, the series up to the first three harmonics is

$$f(x) = \frac{a_0}{2} + (a_1 \cos \theta + b_1 \sin \theta) + (a_2 \cos 2\theta + b_2 \sin 2\theta) + (a_3 \cos 3\theta + b_3 \sin 3\theta)$$

We have the following table (with $N = 6$; $x = 6$ repeats $x = 0$):

x	y	$\theta = \frac{\pi x}{3}$	$\cos \theta$	$\cos 2\theta$	$\cos 3\theta$	$\sin \theta$	$\sin 2\theta$	$\sin 3\theta$	$y \cos \theta$	$y \cos 2\theta$	$y \cos 3\theta$	$y \sin \theta$	$y \sin 2\theta$	$y \sin 3\theta$
0	1.0	0	1	1	1	0	0	0	1	1	1	0	0	0
1	1.4	$\pi/3$	0.5	-0.5	-1	0.866	0.866	0	0.7	-0.7	-1.4	1.2124	1.2124	0
2	1.9	$2\pi/3$	-0.5	-0.5	1	0.866	-0.866	0	-0.95	-0.95	1.9	1.6454	-1.6454	0
3	1.7	π	-1	1	-1	0	0	0	-1.7	1.7	-1.7	0	0	0
4	1.5	$4\pi/3$	-0.5	-0.5	1	-0.866	0.866	0	-0.75	-0.75	1.5	-1.299	1.299	0
5	1.2	$5\pi/3$	0.5	-0.5	-1	-0.866	-0.866	0	0.6	-0.6	-1.2	-1.0392	-1.0392	0
$\sum y = 8.7$									$\sum y \cos \theta = -1.1$	$\sum y \cos 2\theta = -0.3$	$\sum y \cos 3\theta = 0.1$	$\sum y \sin \theta = 0.5196$	$\sum y \sin 2\theta = -0.1732$	$\sum y \sin 3\theta = 0$

Cosine coefficients

$$a_0 = 2 \left(\frac{\sum y}{N} \right) = 2 \left(\frac{8.7}{6} \right) \quad \Rightarrow \quad \boxed{a_0 = 2.9}$$

$$a_1 = 2 \left(\frac{\sum y \cos \theta}{N} \right) = 2 \left(\frac{-1.1}{6} \right) \quad \Rightarrow \quad \boxed{a_1 = -0.366}$$

$$a_2 = 2 \left(\frac{\sum y \cos 2\theta}{N} \right) = 2 \left(\frac{-0.3}{6} \right) \quad \Rightarrow \quad \boxed{a_2 = -0.1}$$

$$a_3 = 2 \left(\frac{\sum y \cos 3\theta}{N} \right) = 2 \left(\frac{0.1}{6} \right) \quad \Rightarrow \quad \boxed{a_3 = 0.033}$$

Sine coefficients

$$b_1 = 2 \left(\frac{\sum y \sin \theta}{N} \right) = 2 \left(\frac{0.5196}{6} \right) \quad \Rightarrow \quad b_1 = 0.1732$$

$$b_2 = 2 \left(\frac{\sum y \sin 2\theta}{N} \right) = 2 \left(\frac{-0.1732}{6} \right) \quad \Rightarrow \quad b_2 = -0.057$$

$$b_3 = 2(0) \quad \Rightarrow \quad b_3 = 0$$

Required Fourier series up to the third harmonic

$$f(\theta) = \frac{2.9}{2} + (-0.366 \cos \theta + 0.1732 \sin \theta) + (-0.1 \cos 2\theta - 0.057 \sin 2\theta) \\ + (0.033 \cos 3\theta + 0), \quad \text{where } \theta = \frac{\pi x}{3}$$

APPLICATION OF FOURIER SERIES

Frequency Response and Oscillating Systems

Overview

A great many engineering devices — a vehicle suspension crossing evenly spaced road joints, a building shaken by rotating machinery, a tuning circuit fed by a switching power supply — are *linear oscillating systems driven by a periodic but non-sinusoidal input*. Such a system cannot be analysed by the elementary “resonance” formula, because the input is not a single sine wave.

Fourier series solves the difficulty completely: the input is resolved into harmonics, each harmonic is passed through the system separately using its **frequency response function**, and the outputs are superposed. The following sections develop that method and apply it to three fully worked problems.

9 Why Fourier Series Enters

Consider the standard damped oscillator driven by an external force $F(t)$:

$$m\ddot{x} + c\dot{x} + kx = F(t), \quad (1)$$

where m is the mass, c the damping constant and k the stiffness. If $F(t) = F_0 \sin \omega t$, every textbook gives the steady-state answer at once. But in practice $F(t)$ is very often periodic and *not* sinusoidal: a square wave from an on/off actuator, a sawtooth from a scanning circuit, a train of triangular pulses from a cam, a half-wave rectified voltage.

Two facts make the general case tractable.

The two enabling facts

- (a) **The system is linear.** Writing (1) as $L[x] = F$ with $L \equiv m \frac{d^2}{dt^2} + c \frac{d}{dt} + k$, we have

$$L[\alpha x_1 + \beta x_2] = \alpha L[x_1] + \beta L[x_2].$$

Hence *responses to separate inputs may simply be added*.

- (b) **Fourier’s theorem.** Any reasonable periodic F of period $T = 2\pi/\omega_0$ can be written as a sum of sinusoids at the frequencies $\omega_0, 2\omega_0, 3\omega_0, \dots$

Put together: resolve the input into harmonics, solve one easy problem per harmonic, add up. This is the whole idea, and it is the single most used application of Fourier series in engineering.

10 Response to One Harmonic: the Frequency Response Function

10.1 Normalised form of the oscillator

Divide (1) by m and introduce

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{undamped natural frequency}), \quad \zeta = \frac{c}{2\sqrt{mk}} \quad (\text{damping ratio}),$$

so that

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F(t)}{m}. \quad (2)$$

For $0 < \zeta < 1$ the complementary function is $e^{-\zeta\omega_nt}(A \cos \omega_d t + B \sin \omega_d t)$, where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Because of the factor $e^{-\zeta\omega_nt}$ this *transient* dies out. What survives is the particular integral, called the **steady-state response**, and it is the only part we compute below.

10.2 The complex-exponential trick

Take the trial input $F(t) = m e^{i\omega t}$ and look for $x = H(i\omega)e^{i\omega t}$. Substituting in (2),

$$(-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2)H(i\omega)e^{i\omega t} = e^{i\omega t},$$

which gives the **frequency response function** (also called the transfer function evaluated on the imaginary axis):

Frequency response function

$$H(i\omega) = \frac{1}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (3)$$

Writing $H(i\omega) = |H(i\omega)| e^{-i\theta(\omega)}$,

$$|H(i\omega)| = \frac{1}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}}, \quad \theta(\omega) = \arctan \frac{2\zeta\omega_n\omega}{\omega_n^2 - \omega^2} \quad (0 \leq \theta \leq \pi). \quad (4)$$

Meaning of H

$|H|$ is the **gain**: how much the system multiplies the amplitude of an input at frequency ω .

θ is the **phase lag**: how far the output sinusoid is delayed behind the input sinusoid.

Consequently, for a real input $F(t) = F_0 \cos \omega t$,

$$x_{ss}(t) = \frac{F_0}{m} |H(i\omega)| \cos(\omega t - \theta(\omega)).$$

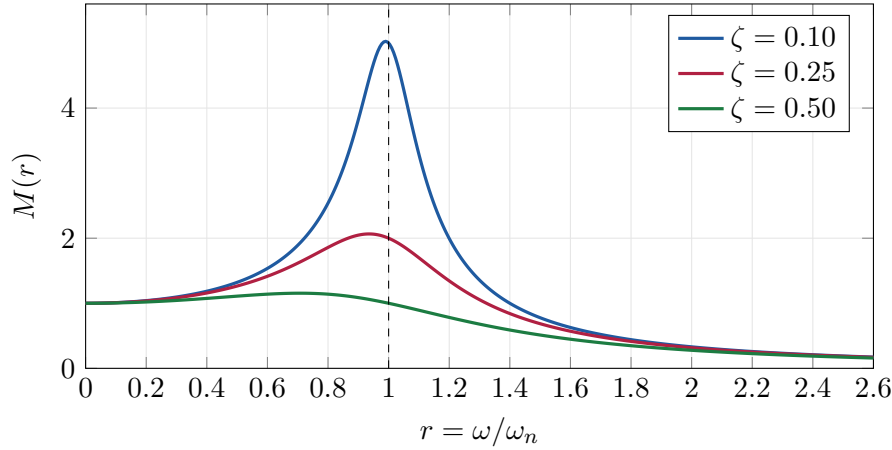
A sinusoid in gives a sinusoid out, at the same frequency, only rescaled and shifted. This is the property that makes the whole method work.

10.3 Magnification factor and resonance curve

It is convenient to use the frequency ratio $r = \omega/\omega_n$ and the static deflection $x_{st} = F_0/k$. Then x_{ss} has amplitude $M(r)x_{st}$, with the **magnification factor**

Magnification factor

$$M(r) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \quad \theta = \arctan \frac{2\zeta r}{1-r^2}. \quad (5)$$



Three features of this curve govern everything that follows.

Features of the resonance curve

- (i) $M \rightarrow 1$ as $r \rightarrow 0$: very slow forcing simply pushes the mass to its static deflection.
- (ii) M peaks sharply near $r = 1$ — exactly at $r = \sqrt{1 - 2\zeta^2}$, with peak value $\frac{1}{2\zeta\sqrt{1 - \zeta^2}}$. Light damping $\Rightarrow$ tall, narrow peak.
- (iii) $M \rightarrow 0$ as $r \rightarrow \infty$: the mass cannot follow very rapid forcing. The oscillator therefore behaves as a **band-pass filter** centred on ω_n .

10.4 Transfer-function notation $G(s)$

In systems and control courses the same information is packaged through the Laplace transform. Taking $\mathcal{L}$ of (1) with zero initial conditions gives $(ms^2 + cs + k)X(s) = F(s)$, so

System transfer function

$$X(s) = G(s)F(s), \quad G(s) = \frac{1}{ms^2 + cs + k} \quad (6)$$

$G(s)$ is the **system transfer function**. Putting $s = j\omega$ (engineers write $j = \sqrt{-1}$) recovers the frequency response of (3): indeed $G(j\omega) = mH(i\omega)$, i.e. $G(j\omega)$ is H with the mass already divided out. The essential statement is the following.

Sinusoidal steady-state theorem

If a stable linear system with transfer function $G(s)$ is driven by $u(t) = A \sin \omega t$, its steady-state output is

$$y_{ss}(t) = A |G(j\omega)| \sin(\omega t + \arg(G(j\omega))). \quad (7)$$

Proof. Write $u = \text{Im}\{Ae^{j\omega t}\}$ and look for $y = \text{Im}\{Ye^{j\omega t}\}$. Substituting in $m\ddot{y} + c\dot{y} + ky = u$,

$$(-m\omega^2 + j\omega c + k)Y = A \implies Y = AG(j\omega) = A|G(j\omega)|e^{j \arg G(j\omega)},$$

and taking the imaginary part of $Y e^{j\omega t}$ gives (7). □

Note that $\arg G(j\omega)$ is negative for a stable second-order system, so (7) expresses a lag, in agreement with (4): $\arg G(j\omega) = -\theta(\omega)$. Combined with a Fourier series this is again the master result (10), only written in control-engineering notation — Problem 3 below works entirely in this language.

11 The Fourier-Series Method

Let F be periodic of period T , with fundamental angular frequency $\omega_0 = 2\pi/T$, and let its Fourier series be

$$F(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t), \quad (8)$$

$$a_n = \frac{2}{T} \int_0^T F(t) \cos n\omega_0 t \, dt, \quad b_n = \frac{2}{T} \int_0^T F(t) \sin n\omega_0 t \, dt.$$

It is often neater to write (8) in **amplitude–phase form**,

$$F(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t - \psi_n), \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \tan \psi_n = \frac{b_n}{a_n}. \quad (9)$$

Master result: steady-state response to periodic forcing

With H as in (3) and the n -th harmonic frequency $n\omega_0$,

$$x_{ss}(t) = \underbrace{\frac{a_0}{2k}}_{\text{mean shift}} + \sum_{n=1}^{\infty} \frac{c_n}{m} |H(in\omega_0)| \cos(n\omega_0 t - \psi_n - \theta(n\omega_0)). \quad (10)$$

Equivalently, in $\{a_n, b_n\}$ form,

$$x_{ss}(t) = \frac{a_0}{2k} + \sum_{n=1}^{\infty} \frac{(k - mn^2\omega_0^2)(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) + cn\omega_0(a_n \sin n\omega_0 t - b_n \cos n\omega_0 t)}{(k - mn^2\omega_0^2)^2 + (cn\omega_0)^2}.$$

11.1 Working recipe

Step-by-step procedure

1. Identify m, c, k ; compute $\omega_n = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{mk})$.
2. Find the period T of F , hence $\omega_0 = 2\pi/T$, and obtain a_0, a_n, b_n .
3. List the harmonic frequencies $\omega_0, 2\omega_0, 3\omega_0, \dots$ and locate the one nearest ω_n — **that harmonic will dominate**.
4. Compute the gain $|H(in\omega_0)|$ and the phase $\theta(n\omega_0)$ for each harmonic.
5. Assemble (10); keep only the terms that are numerically significant.

11.2 Why so few terms are needed

The n -th output amplitude is $c_n |H(in\omega_0)|/m$, a product of two decaying factors: the Fourier coefficients c_n themselves decrease (like $1/n$ for a jump discontinuity, like $1/n^2$ for a corner), and the gain

$|H|$ falls off like $1/(n\omega_0)^2$ for $n\omega_0 \gg \omega_n$. The response is therefore *much smoother than the input*: an oscillator fed a square wave does not move in square-wave fashion but in an almost sinusoidal fashion. Mechanical systems are natural low-pass/band-pass filters.

11.3 Resonance with a non-sinusoidal input

The key engineering warning

Resonance does *not* require the forcing to be sinusoidal, and it does *not* require the driving period to match the natural period. It is enough that **some harmonic** $n\omega_0$ falls close to ω_n . A machine running at a perfectly safe fundamental speed can still shake itself apart through its 5th or 7th harmonic.

Two practical consequences:

- If the input has only odd harmonics (as a square wave does, and any half-wave symmetric input does), then an even-numbered coincidence $n\omega_0 = \omega_n$ is harmless — that harmonic is simply absent.
- Damping controls the disaster. With ζ small, the resonant term is amplified by roughly $1/(2\zeta)$ relative to its static value, so the response becomes nearly a pure sinusoid at ω_n regardless of the shape of the input. The system *selects* one harmonic — exactly the principle used deliberately in a radio tuning circuit.

11.4 Mean-square response (Parseval)

Because distinct harmonics are orthogonal over a period, the mean square of the response is the sum of the mean squares of its harmonics.

Parseval for the response

$$\overline{x^2} = \left(\frac{a_0}{2k}\right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} A_n^2, \quad A_n = \frac{c_n}{m} |H(in\omega_0)|. \quad (11)$$

This gives the RMS displacement (or RMS current, in the electrical case) directly from the harmonic amplitudes, and shows at a glance what fraction of the energy sits in the resonant harmonic.

12 The Electrical Analogue

A series *RLC* circuit driven by an EMF $E(t)$ obeys

$$L\ddot{q} + R\dot{q} + \frac{q}{C} = E(t), \quad i = \dot{q}, \quad (12)$$

which is (1) with the correspondences below.

Mechanical	Electrical	
mass m	inductance L	inertia
damping c	resistance R	dissipation
stiffness k	elastance $1/C$	restoring
force $F(t)$	EMF $E(t)$	input
displacement x	charge q	output
velocity $\dot{x}$	current i	
$\omega_n = \sqrt{k/m}$	$\omega_r = 1/\sqrt{LC}$	resonant frequency

For the *current* the algebra is even cleaner. For an input $E_n \sin n\omega_0 t$ the steady-state current is

Harmonic current and impedance

$$i_n(t) = \frac{E_n}{|Z_n|} \sin(n\omega_0 t - \phi_n), \quad Z_n = R + i \underbrace{\left(n\omega_0 L - \frac{1}{n\omega_0 C} \right)}_{X_n}, \quad (13)$$

$$|Z_n| = \sqrt{R^2 + X_n^2}, \quad \phi_n = \arctan \frac{X_n}{R}.$$

The harmonic for which $X_n = 0$, i.e. $n\omega_0 = 1/\sqrt{LC}$, sees a purely resistive impedance $|Z| = R$; it is passed with maximum amplitude and with zero phase shift. All others are attenuated. The sharpness of this selection is measured by the quality factor $Q = \omega_r L/R$.

13 Worked Problems on Frequency Response

13.1 Problem 1 : Square-wave forcing of a mass–spring–damper

Problem 1

A machine part of mass $m = 1$ kg is mounted on a spring of stiffness $k = 25$ N/m with a dashpot of constant $c = 0.1$ Ns/m. It is driven by a solenoid that switches direction every π seconds, producing the square-wave force

$$F(t) = \begin{cases} +1 \text{ N}, & 0 < t < \pi, \\ -1 \text{ N}, & \pi < t < 2\pi, \end{cases} \quad F(t + 2\pi) = F(t).$$

Find the steady-state displacement. Identify the dominant harmonic, give the first few terms numerically, and compute the RMS displacement.

Solution.

Step 1 : System parameters

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{25} = 5 \text{ rad/s}, \quad \zeta = \frac{c}{2\sqrt{mk}} = \frac{0.1}{2 \cdot 5} = 0.01.$$

The damping is very light ($\zeta = 1\%$), so the resonance peak is tall and narrow. The governing equation is

$$\ddot{x} + 0.1 \dot{x} + 25x = F(t).$$

Step 2 : Fourier series of the square wave

Here $T = 2\pi$, so $\omega_0 = 1$ rad/s. The function is **odd**, hence $a_0 = a_n = 0$ and only sine terms occur:

$$b_n = \frac{2}{T} \int_0^T F(t) \sin nt \, dt = \frac{2}{2\pi} \left[\int_0^\pi \sin nt \, dt - \int_\pi^{2\pi} \sin nt \, dt \right].$$

Now

$$\int_0^\pi \sin nt \, dt = \left[\frac{-\cos nt}{n} \right]_0^\pi = \frac{1 - \cos n\pi}{n}, \quad \int_\pi^{2\pi} \sin nt \, dt = \left[\frac{-\cos nt}{n} \right]_\pi^{2\pi} = \frac{\cos n\pi - 1}{n},$$

so

$$b_n = \frac{1}{\pi} \cdot \frac{2(1 - \cos n\pi)}{n} = \frac{2(1 - (-1)^n)}{n\pi} = \begin{cases} \frac{4}{n\pi}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Therefore

Fourier series of the square wave

$$F(t) = \frac{4}{\pi} \left(\sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \frac{\sin 7t}{7} + \dots \right) = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\sin nt}{n}. \quad (14)$$

Step 3 : Spot the resonance before computing anything

The harmonic frequencies present are $1, 3, 5, 7, 9, \dots$ rad/s (odd only). Since $\omega_n = 5$, the **fifth harmonic sits exactly on resonance**. Note that the driving fundamental $\omega_0 = 1$ is nowhere near $\omega_n = 5$ — a naive check of the fundamental alone would have declared the machine safe.

Step 4 : Response to a single harmonic

Solve $\ddot{x} + 0.1\dot{x} + 25x = B_n \sin nt$ with $B_n = 4/(n\pi)$. Try $x_n = P_n \cos nt + Q_n \sin nt$. Then

$$\dot{x}_n = n(-P_n \sin nt + Q_n \cos nt), \quad \ddot{x}_n = -n^2(P_n \cos nt + Q_n \sin nt),$$

so

$$(25 - n^2)(P_n \cos nt + Q_n \sin nt) + 0.1n(-P_n \sin nt + Q_n \cos nt) = B_n \sin nt.$$

Comparing coefficients:

$$\cos nt : (25 - n^2)P_n + 0.1n Q_n = 0, \quad \sin nt : (25 - n^2)Q_n - 0.1n P_n = B_n.$$

Solving this 2×2 system, with $\Delta_n = (25 - n^2)^2 + (0.1n)^2$,

$$P_n = -\frac{0.1n B_n}{\Delta_n}, \quad Q_n = \frac{(25 - n^2)B_n}{\Delta_n}$$

and the amplitude of the n -th harmonic of the response is

$$A_n = \sqrt{P_n^2 + Q_n^2} = \frac{B_n}{\sqrt{\Delta_n}} = \frac{4}{n\pi \sqrt{(25 - n^2)^2 + 0.01n^2}}.$$

Step 5 : The resonant term $n = 5$, done separately

Here $25 - n^2 = 0$, so the formulae of Step 4 collapse:

$$Q_5 = 0, \quad P_5 = -\frac{0.1 \cdot 5 \cdot B_5}{(0.1 \cdot 5)^2} = -\frac{0.5 B_5}{0.25} = -2B_5 = -\frac{8}{5\pi} = -0.50930.$$

Hence $x_5(t) = -0.50930 \cos 5t$, i.e. amplitude 0.5093 m lagging the input $\sin 5t$ by exactly 90° — the signature of exact resonance.

Step 6 : Numerical table

n	$B_n = \frac{4}{n\pi}$	Δ_n	P_n	Q_n	A_n
1	1.27324	576.01	-0.000221	0.053051	0.053051
3	0.42441	256.09	-0.000497	0.026517	0.026521
5	0.25465	0.25	-0.509296	0	0.509296
7	0.18189	576.49	-0.000221	-0.007572	0.007576
9	0.14147	3136.81	-0.000041	-0.002526	0.002526
11	0.11575	9217.21	-0.000014	-0.001206	0.001206

Step 7 : The answer

$$\begin{aligned} x_{ss}(t) = & 0.053051 \sin t - 0.000221 \cos t \\ & + 0.026517 \sin 3t - 0.000497 \cos 3t \\ & - 0.509296 \cos 5t \\ & - 0.007572 \sin 7t - 0.000221 \cos 7t \\ & - 0.002526 \sin 9t - 0.000041 \cos 9t + \dots \end{aligned}$$

or, keeping only the visible terms,

Steady-state displacement

$$x_{ss}(t) \approx -0.5093 \cos 5t + 0.0531 \sin t + 0.0265 \sin 3t - 0.0076 \sin 7t \quad (\text{m})$$

Step 8 : RMS displacement

By (11),

$$x_{\text{rms}} = \sqrt{\frac{1}{2} \sum A_n^2} = \sqrt{\frac{1}{2} (0.053051^2 + 0.026521^2 + 0.509296^2 + 0.007576^2 + \dots)} = 0.3626 \text{ m}.$$

The $n = 5$ term alone accounts for $0.5093^2 / \sum A_n^2 = 98.6\%$ of the mean-square displacement.

Discussion

- The resonant harmonic is about **10 times** the fundamental (0.5093 vs. 0.0531) even though the *input* fundamental is 5 times *larger* than the input fifth harmonic. The system has amplified a component of the force that was, in the input, comparatively minor.
- The response is essentially a clean sinusoid at 5 rad/s: the square wave has been filtered into

a near-pure tone. The exact peak displacement is 0.573 m (slightly above 0.5093 because the small non-resonant terms occasionally add in phase).

- A useful shortcut visible in the table: since $nB_n = 4/\pi$ is independent of n , we get $P_n = -0.1(4/\pi)/\Delta_n = -0.12732/\Delta_n$ for every odd n .
- **Design fix.** To protect the machine one may either raise c (increase ζ , flattening the peak), or change k or m so that ω_n falls between two odd multiples of ω_0 — e.g. $\omega_n = 4$ or $\omega_n = 6$ would place no harmonic on resonance at all, since even harmonics are absent from a square wave.

13.2 Problem 2 : Sawtooth EMF applied to a series RLC circuit

Problem 2

A series circuit has $L = 1$ H, $R = 0.5 \Omega$ and $C = \frac{1}{9}$ F. It is driven by the sawtooth EMF

$$E(t) = \frac{10t}{\pi} \text{ volts for } -\pi < t < \pi, \quad E(t + 2\pi) = E(t),$$

(peak value 10 V, period 2π s). Find the steady-state current $i(t)$, identify the harmonic the circuit selects, and find the RMS current.

Solution.

Step 1 : Circuit parameters

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \cdot \frac{1}{9}}} = 3 \text{ rad/s}, \quad Q = \frac{\omega_r L}{R} = \frac{3 \times 1}{0.5} = 6.$$

The circuit is moderately sharply tuned to 3 rad/s. The governing equation is

$$\ddot{q} + 0.5\dot{q} + 9q = E(t), \quad i = \dot{q}.$$

Step 2 : Fourier series of the sawtooth

$T = 2\pi$, so $\omega_0 = 1$ rad/s. Since $E(t) = 10t/\pi$ is **odd** on $(-\pi, \pi)$, $a_0 = a_n = 0$ and

$$b_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} \frac{10t}{\pi} \sin nt \, dt = \frac{2}{\pi} \cdot \frac{10}{\pi} \int_0^{\pi} t \sin nt \, dt.$$

Integrating by parts,

$$\int_0^{\pi} t \sin nt \, dt = \left[-\frac{t \cos nt}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nt \, dt = -\frac{\pi \cos n\pi}{n} + 0 = \frac{\pi(-1)^{n+1}}{n}.$$

Hence

$$b_n = \frac{20}{\pi^2} \cdot \frac{\pi(-1)^{n+1}}{n} = \frac{20(-1)^{n+1}}{n\pi},$$

Fourier series of the sawtooth

$$E(t) = \frac{20}{\pi} \left(\sin t - \frac{\sin 2t}{2} + \frac{\sin 3t}{3} - \frac{\sin 4t}{4} + \cdots \right) = \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nt. \quad (15)$$

Unlike the square wave, the sawtooth contains *all* harmonics, even and odd.

Step 3 : Impedance of each harmonic

With $\omega_0 = 1$, harmonic n has frequency n rad/s, so from (13)

$$X_n = nL - \frac{1}{nC} = n - \frac{9}{n}, \quad |Z_n| = \sqrt{0.25 + \left(n - \frac{9}{n}\right)^2}, \quad \phi_n = \arctan \frac{X_n}{0.5}.$$

$X_n = 0$ when $n^2 = 9$, i.e. $n = 3$: **the third harmonic is at resonance**, sees $|Z_3| = R = 0.5 \Omega$ only, and suffers no phase shift.

Step 4 : Harmonic-by-harmonic current

For the input term $E_n \sin nt$ with $E_n = \frac{20(-1)^{n+1}}{n\pi}$,

$$i_n(t) = \frac{E_n}{|Z_n|} \sin(nt - \phi_n).$$

n	E_n (V)	$X_n = n - \frac{9}{n}$	$ Z_n $ (Ω)	ϕ_n	$ E_n / Z_n $ (A)
1	6.3662	-8.0000	8.0156	-86.42°	0.7942
2	-3.1831	-2.5000	2.5495	-78.69°	1.2485
3	2.1221	0	0.5000	0°	4.2441
4	-1.5916	1.7500	1.8200	74.05°	0.8745
5	1.2732	3.2000	3.2388	81.12°	0.3931
6	-1.0610	4.5000	4.5277	83.66°	0.2343
7	0.9095	5.7143	5.7361	85.00°	0.1586

Step 5 : The answer

$$\begin{aligned} i(t) = & 0.7942 \sin(t + 86.42^\circ) - 1.2485 \sin(2t + 78.69^\circ) \\ & + 4.2441 \sin 3t - 0.8745 \sin(4t - 74.05^\circ) \\ & + 0.3931 \sin(5t - 81.12^\circ) - 0.2343 \sin(6t - 83.66^\circ) + \dots \end{aligned}$$

so that, to a good approximation,

Steady-state current

$$i(t) \approx 4.244 \sin 3t \quad (\text{A})$$

Step 6 : RMS current

By Parseval,

$$i_{\text{rms}} = \sqrt{\frac{1}{2} \sum_{n \geq 1} \left(\frac{|E_n|}{|Z_n|} \right)^2} = \sqrt{\frac{1}{2} (0.7942^2 + 1.2485^2 + 4.2441^2 + 0.8745^2 + \dots)} = 3.259 \text{ A},$$

of which the third harmonic supplies $4.2441^2 / \sum (\cdot)^2 = 84.8\%$ of the mean square.

Discussion

- The input voltage is a sawtooth whose *largest* harmonic is the fundamental (6.37 V); the third harmonic is only 2.12 V. Yet the output current is dominated by the third harmonic, because $|Z_3| = 0.5$ while $|Z_1| = 8.02$ — a factor of 16 in favour of $n = 3$.
- **This is exactly how a tuned circuit works.** Feed it a signal rich in harmonics and it returns, to good accuracy, a single sinusoid at its own resonant frequency. Increasing Q (smaller R , or larger L/C) sharpens the selection; the “purity” of the output improves as $R \rightarrow 0$.
- Note the phase behaviour along the table: below resonance $X_n < 0$ (capacitive), the current *leads*; at resonance the phase is zero; above resonance $X_n > 0$ (inductive), the current *lags*, tending to 90° for large n . The phase swings through 180° across resonance, and does so faster the higher the Q .
- If instead the *charge* $q(t)$ were required, one would divide each current harmonic by $n\omega_0$ and shift its phase by 90° ; the higher harmonics would then be even more strongly suppressed.

13.3 Problem 3 : Transfer-function form – a general proof, then numbers**Problem 3**

The mass–spring–damper system of Fig. 1(a) is initially at rest in its equilibrium position when the mass is subjected to the externally applied periodic force $P(t)$ of Fig. 1(b): a square wave of amplitude 10 N and period 2π s. The displacement $x(t)$ satisfies

$$M \frac{d^2x}{dt^2} + B \frac{dx}{dt} + Kx = P(t), \quad G(s) = \frac{1}{Ms^2 + Bs + K},$$

with $M = 10$ kg, $B = 0.5$ kg s $^{-1}$, $K = 250$ N m $^{-1}$.

(i) Prove that the steady-state response of the system to the n -th harmonic is

$$x_{ssn}(t) = \frac{40}{\pi(2n-1)} \left| G(j(2n-1)) \right| \sin \left[(2n-1)t + \arg(G(j(2n-1))) \right].$$

(ii) Evaluate the first few harmonics numerically and identify the dominant one.

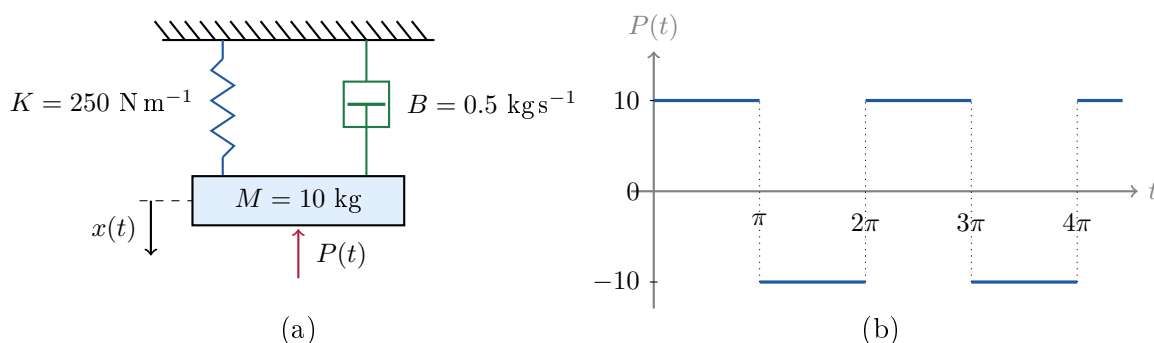


Figure 1: (a) The mass–spring–damper system. (b) The applied periodic force $P(t)$: a square wave of amplitude 10 N and period 2π s.

Solution.

Part (i) : the proof**Step 1 : Fourier series of $P(t)$**

From Fig. 1(b), P has period $T = 2\pi$ (so $\omega_0 = 2\pi/T = 1$ rad/s) and

$$P(t) = \begin{cases} +10, & 0 < t < \pi, \\ -10, & \pi < t < 2\pi. \end{cases}$$

P is an odd function, so $a_0 = 0$ and $a_m = 0$ for all m . For the sine coefficients,

$$b_m = \frac{2}{T} \int_0^T P(t) \sin mt \, dt = \frac{1}{\pi} \left[\int_0^\pi 10 \sin mt \, dt - \int_\pi^{2\pi} 10 \sin mt \, dt \right].$$

Since

$$\int_0^\pi \sin mt \, dt = \frac{1 - (-1)^m}{m}, \quad \int_\pi^{2\pi} \sin mt \, dt = \frac{(-1)^m - 1}{m},$$

we get

$$b_m = \frac{10}{\pi} \cdot \frac{2(1 - (-1)^m)}{m} = \begin{cases} \frac{40}{m\pi}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases}$$

Only odd m survive, so we index them by $n = 1, 2, 3, \dots$ through $m = 2n - 1$:

Fourier series of $P(t)$

$$P(t) = \sum_{n=1}^{\infty} \frac{40}{\pi(2n-1)} \sin[(2n-1)t] \quad (16)$$

Thus the n -th harmonic of the input is

$$P_n(t) = A_n \sin \Omega_n t, \quad A_n = \frac{40}{\pi(2n-1)}, \quad \Omega_n = (2n-1) \text{ rad/s}.$$

(We write the harmonic frequency as Ω_n so that it is not confused with the natural frequency ω_n of the system.)

Step 2 : Steady-state response to that single harmonic

By linearity we may drive the system with P_n alone. Using the sinusoidal steady-state theorem (7) — reproduced here in full for completeness — we seek $x_{ssn} = \text{Im}\{X e^{j\Omega_n t}\}$ corresponding to $P_n = \text{Im}\{A_n e^{j\Omega_n t}\}$. Substituting into $M\ddot{x} + B\dot{x} + Kx = P_n$ and cancelling $e^{j\Omega_n t}$,

$$(M(j\Omega_n)^2 + B(j\Omega_n) + K)X = A_n \implies X = \frac{A_n}{M(j\Omega_n)^2 + B(j\Omega_n) + K} = A_n G(j\Omega_n).$$

Writing $G(j\Omega_n) = |G(j\Omega_n)| e^{j \arg G(j\Omega_n)}$,

$$\begin{aligned} x_{ssn}(t) &= \text{Im}\left\{A_n |G(j\Omega_n)| e^{j(\Omega_n t + \arg G(j\Omega_n))}\right\} \\ &= A_n |G(j\Omega_n)| \sin(\Omega_n t + \arg G(j\Omega_n)). \end{aligned}$$

Step 3 : Substitute A_n and Ω_n from Step 1

Required result

$$x_{ssn}(t) = \frac{40}{\pi(2n-1)} \left| G(j(2n-1)) \right| \sin \left[(2n-1)t + \arg(G(j(2n-1))) \right] \quad (17)$$

■

By superposition the complete steady-state motion is $x_{ss}(t) = \sum_{n \geq 1} x_{ssn}(t)$; the transient omitted here decays like $e^{-\zeta\omega_n t}$, which is legitimate because the system is stable.

Part (ii) : the numbers**Step 4 : System constants**

$$\omega_n = \sqrt{\frac{K}{M}} = \sqrt{\frac{250}{10}} = 5 \text{ rad/s}, \quad \zeta = \frac{B}{2\sqrt{MK}} = \frac{0.5}{2\sqrt{2500}} = \frac{0.5}{100} = 0.005.$$

The damping ratio is a mere 0.5%, so $Q \approx 1/(2\zeta) = 100$: an extremely sharply tuned system. The harmonics present are $\omega = 1, 3, 5, 7, 9, \dots$ rad/s, and $\Omega_n = \omega_n = 5$ occurs at $2n-1 = 5$, i.e. **at** $n = 3$.

Step 5 : Evaluate $G(j\omega)$

With $M = 10$, $B = 0.5$, $K = 250$,

$$G(j\omega) = \frac{1}{K - M\omega^2 + jB\omega} = \frac{1}{250 - 10\omega^2 + 0.5j\omega},$$

$$|G(j\omega)| = \frac{1}{\sqrt{(250 - 10\omega^2)^2 + (0.5\omega)^2}}, \quad \arg G(j\omega) = -\arctan \frac{0.5\omega}{250 - 10\omega^2}$$

(the arctangent taken in the correct quadrant: the argument lies in $(-180^\circ, 0)$ throughout).

Step 6 : Numerical table

n	$\omega = 2n-1$	$A_n = \frac{40}{\pi(2n-1)}$	$K - M\omega^2$	$B\omega$	$ G(j\omega) $	$A_n G $ (m)
1	1	12.7324	240	0.5	0.0041667	0.053052
2	3	4.2441	160	1.5	0.0062497	0.026525
3	5	2.5465	0	2.5	0.4000000	1.018592
4	7	1.8189	-240	3.5	0.0041662	0.007578
5	9	1.4147	-560	4.5	0.0017857	0.002526
6	11	1.1575	-960	5.5	0.0010417	0.001206

n	$\arg G(j\omega)$	$x_{ssn}(t)$
1	-0.119°	$0.05305 \sin(t - 0.119^\circ)$
2	-0.537°	$0.02652 \sin(3t - 0.537^\circ)$
3	-90°	$1.01859 \sin(5t - 90^\circ) = -1.01859 \cos 5t$
4	-179.164°	$0.00758 \sin(7t - 179.164^\circ)$
5	-179.540°	$0.00253 \sin(9t - 179.540^\circ)$
6	-179.672°	$0.00121 \sin(11t - 179.672^\circ)$

Step 7 : Result**Steady-state displacement**

$$x_{ss}(t) \approx -1.0186 \cos 5t + 0.0531 \sin t + 0.0265 \sin 3t - 0.0076 \sin 7t + \cdots \quad (\text{m})$$

The RMS displacement is $\sqrt{\frac{1}{2} \sum (A_n |G|)^2} = 0.7215 \text{ m}$, of which the $n = 3$ harmonic supplies 99.7% of the mean square. The exact peak excursion is 1.083 m.

Discussion

- **Why $n = 3$ gives exactly -90° .** At $\omega = \omega_n$ the real part $K - M\omega^2$ vanishes, so $G(j\omega_n) = 1/(jB\omega_n) = -j/(B\omega_n)$: purely imaginary and negative, i.e. $|G| = 1/(B\omega_n) = 1/2.5 = 0.4$ and $\arg G = -90^\circ$. *At resonance the response always lags the force by a quarter cycle, whatever the damping* — and its amplitude is controlled by B alone. This is the cleanest single diagnostic for resonance in the laboratory.
- **Note the sign flip past resonance.** For $\omega < \omega_n$ the phase is near 0° (mass moves with the force, stiffness-controlled); for $\omega > \omega_n$ it is near -180° (mass moves against the force, mass-controlled). The whole 180° swing happens between $\omega = 3$ and $\omega = 7$.
- **Comparison with Problem 1.** Dividing this equation by $M = 10$ gives $\ddot{x} + 0.05\dot{x} + 25x = P(t)/10$, which is Problem 1 with the *damping halved* ($\zeta = 0.005$ instead of 0.01). Sure enough, the resonant amplitude has exactly doubled: $1.0186 = 2 \times 0.5093$, while every non-resonant harmonic is essentially unchanged. This illustrates the rule (resonant amplitude) $\propto 1/\zeta$.
- **Practical reading.** A 10 N force gives a static deflection of only $10/250 = 0.04 \text{ m}$, yet the steady-state motion has an amplitude of over 1 m — a magnification of about 25 over the static value, produced entirely by a harmonic the designer may never have thought about. Adding damping, or shifting ω_n off the odd integers, is essential.

14 Summary of the Application

Formulae to remember

System: $m\ddot{x} + c\dot{x} + kx = F(t), \quad \omega_n = \sqrt{k/m}, \quad \zeta = \frac{c}{2\sqrt{mk}}$

Frequency response: $H(i\omega) = \frac{1}{m[(\omega_n^2 - \omega^2) + 2i\zeta\omega_n\omega]}$

Transfer function: $G(s) = \frac{1}{ms^2 + cs + k}; \quad A \sin \omega t \mapsto A|G(j\omega)| \sin(\omega t + \arg G(j\omega))$

Gain / phase: $|H| = \frac{1/m}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}}, \quad \theta = \arctan \frac{2\zeta\omega_n\omega}{\omega_n^2 - \omega^2}$

Input: $F(t) = \frac{a_0}{2} + \sum c_n \cos(n\omega_0 t - \psi_n), \quad \omega_0 = 2\pi/T$

Output: $x_{ss}(t) = \frac{a_0}{2k} + \sum_{n \geq 1} c_n |H(in\omega_0)| \cos(n\omega_0 t - \psi_n - \theta(n\omega_0))$

RLC current: $i_n = \frac{E_n}{|Z_n|} \sin(n\omega_0 t - \phi_n), \quad Z_n = R + i\left(n\omega_0 L - \frac{1}{n\omega_0 C}\right)$

Parseval: $\overline{x^2} = \left(\frac{a_0}{2k}\right)^2 + \frac{1}{2} \sum A_n^2$

The three ideas in one sentence each

1. *Linearity* lets us handle one harmonic at a time and add.
2. *Fourier series* supplies the harmonics of any periodic input.
3. *The frequency response function* tells us, harmonic by harmonic, how much the system amplifies and how much it delays — and any harmonic that lands near ω_n is amplified enormously.

Practice Exercises

Practice Exercises

1. For $\ddot{x} + 0.2\dot{x} + 16x = F(t)$ with F the square wave of Problem 1, show that no harmonic is resonant, and explain why the response is now nearly proportional to the input. (*Hint: $\omega_n = 4$ is even; the square wave has only odd harmonics.*)
2. A triangular force $F(t) = |t|$ on $-\pi < t < \pi$, period 2π , drives $\ddot{x} + 0.1\dot{x} + 9x = F(t)$. Obtain the Fourier series $F(t) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\cos nt}{n^2}$ and find the steady-state response, including the constant term. Which harmonic dominates?
3. Repeat Problem 2 with $R = 2\Omega$. Show that $Q = 1.5$ and that the third harmonic now supplies only about a third of the mean-square current — quantifying how damping destroys selectivity.
4. A half-wave rectified voltage $E(t) = E_0 \sin \omega_0 t$ for $0 < t < \pi/\omega_0$ and 0 for $\pi/\omega_0 < t < 2\pi/\omega_0$ is applied to a series RLC circuit. Find its Fourier series and determine the DC component of the resulting current. (*Hint: at $\omega = 0$ a capacitor is an open circuit — interpret your answer.*)
5. Show that for $\zeta \ll 1$ the peak of $M(r)$ occurs at $r = \sqrt{1 - 2\zeta^2}$ with value $\frac{1}{2\zeta\sqrt{1 - \zeta^2}} \approx \frac{1}{2\zeta}$, and deduce the half-power bandwidth $\Delta r \approx 2\zeta$.

TWO MARKS QUESTIONS AND ANSWERS

Two Marks Questions and Answers

Coverage

The twenty questions below span the whole of Unit II: Dirichlet's conditions, Euler's formulae, even and odd functions, waveform diagrams, change of interval, behaviour at a point of discontinuity, half range series, Parseval's identity and RMS value, the complex form, harmonic analysis, and the frequency-response application.

Q1. Dirichlet's conditions

Q1

State Dirichlet's conditions for the Fourier expansion of a function $f(x)$.

Answer. A function $f(x)$ defined in $(\alpha, \alpha + 2\pi)$ can be expanded in a Fourier series if

- (i) $f(x)$ is **single valued**;
- (ii) $f(x)$ has a **finite number of maxima and minima**;
- (iii) $f(x)$ has a **finite number of discontinuities**;
- (iv) $f(x)$ is **continuous or piecewise continuous**, and periodic with $f(x + 2\pi) = f(x)$.

Q2. Euler's formulae

Q2

Write the Fourier series of $f(x)$ in $(0, 2\pi)$ together with Euler's formulae for its coefficients.

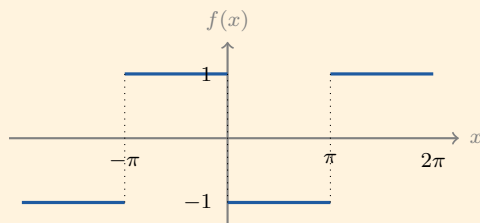
Answer.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Q3. Square wave – identify a_0 and a_n **Q3**

The periodic function shown below has period 2π . Without performing any integration, write down the values of a_0 and a_n , and state which coefficients survive.



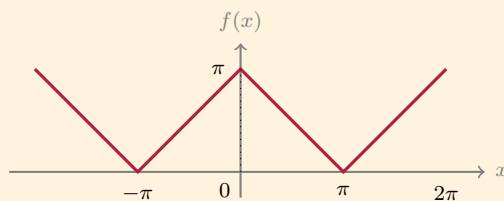
Answer. From the graph, $f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases}$ so that $f(-x) = -f(x)$, i.e. $f(x)$ is an **odd function**. Hence

$$a_0 = 0, \quad a_n = 0, \quad \text{only } b_n \text{ survive}$$

and the series is a pure sine series, $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$ with $b_n = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = \frac{4}{n\pi}$ for n odd and 0 for n even.

Q4. Triangular wave – identify b_n and find a_0 **Q4**

For the triangular wave shown, of period 2π , state the value of b_n and evaluate a_0 .



Answer. The graph is that of $f(x) = |x|$ in $(-\pi, \pi)$, and $f(-x) = |-x| = |x| = f(x)$, so $f(x)$ is an **even function**. Therefore $b_n = 0$, and

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} x \, dx = \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi} = \frac{2}{\pi} \cdot \frac{\pi^2}{2}$$

$$b_n = 0, \quad a_0 = \pi$$

Q5. Odd function – vanishing coefficients**Q5**

If $f(x) = x^3$ in $(-\pi, \pi)$, which of a_0 , a_n , b_n vanish, and why?

Answer. $f(-x) = (-x)^3 = -x^3 = -f(x)$, so $f(x)$ is an **odd function**. The product $f(x) \cos nx$ is then odd and integrates to zero over the symmetric interval $(-\pi, \pi)$, whereas $f(x) \sin nx$ is even. Hence

$$a_0 = 0, \quad a_n = 0, \quad b_n = \frac{2}{\pi} \int_0^\pi x^3 \sin nx \, dx \neq 0$$

Q6. Constant term for $f(x) = x^2$

Q6

Find a_0 in the Fourier expansion of $f(x) = x^2$ in $(-\pi, \pi)$.

Answer. $f(x) = x^2$ is even, so

$$a_0 = \frac{2}{\pi} \int_0^\pi x^2 \, dx = \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^\pi = \frac{2}{\pi} \cdot \frac{\pi^3}{3}$$

$$a_0 = \frac{2\pi^2}{3}$$

Q7. Half range sine coefficients

Q7

Find b_n in the half range sine series of $f(x) = 1$ in $(0, \pi)$.

Answer.

$$b_n = \frac{2}{\pi} \int_0^\pi 1 \cdot \sin nx \, dx = \frac{2}{\pi} \left(\frac{-\cos nx}{n} \right)_0^\pi = \frac{2}{n\pi} (1 - \cos n\pi) = \frac{2}{n\pi} (1 - (-1)^n)$$

$$b_n = \begin{cases} \frac{4}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

Q8. Half range cosine series of $f(x) = x$

Q8

Find a_0 in the half range cosine series of $f(x) = x$ in $(0, \pi)$.

Answer.

$$a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^\pi = \frac{2}{\pi} \cdot \frac{\pi^2}{2}$$

$$a_0 = \pi$$

Q9. Change of length of interval

Q9

A function $f(x)$ is defined in $(0, 4)$ and is periodic of period 4. Write the form of its Fourier series and the formula for a_n .

Answer. Here $2\ell = 4$, so $\ell = 2$, and

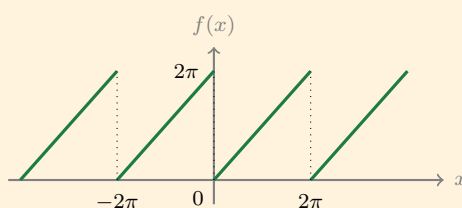
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{2},$$

$$a_n = \frac{1}{\ell} \int_0^{2\ell} f(x) \cos \frac{n\pi x}{\ell} dx = \frac{1}{2} \int_0^4 f(x) \cos \frac{n\pi x}{2} dx$$

Q10. Sawtooth wave – value at a point of discontinuity

Q10

The sawtooth function $f(x) = x$, $0 < x < 2\pi$, is extended periodically as shown. To what value does its Fourier series converge at $x = 0$?



Answer. $x = 0$ is a **point of discontinuity**. There the series converges to the average of the left- and right-hand limits:

$$(f(x))_{x=0} = \frac{f(0^+) + f(0^-)}{2} = \frac{0 + 2\pi}{2}$$

$$(f(x))_{x=0} = \pi$$

Q11. Complex form of the Fourier series

Q11

Write the complex (exponential) form of the Fourier series of $f(x)$ in $(-\pi, \pi)$ and the formula for C_n .

Answer.

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}, \quad C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

More generally, in an interval of length 2ℓ , $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/\ell}$ with $C_n = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-in\pi x/\ell} dx$.

Q12. Parseval's identity

Q12

State Parseval's identity for $f(x)$ defined in $(-\pi, \pi)$, and write the form it takes when $f(x)$ is an even function.

Answer. For $f(x)$ in $(-\pi, \pi)$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} (f(x))^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

If $f(x)$ is **even**, then $b_n = 0$ and the identity reduces to

$$\frac{1}{\pi} \int_0^{\pi} (f(x))^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2$$

Q13. RMS value from the definition

Q13

Define the RMS value of $y = f(x)$ in (a, b) , and hence find the RMS value of $f(x) = x$ in $(0, 2)$.

Answer. By definition $\bar{y}^2 = \frac{1}{b-a} \int_a^b (f(x))^2 dx$. Here

$$\bar{y}^2 = \frac{1}{2-0} \int_0^2 x^2 dx = \frac{1}{2} \left(\frac{x^3}{3} \right)_0^2 = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}$$

$$\bar{y} = \frac{2}{\sqrt{3}} \approx 1.1547$$

Q14. RMS value through Parseval's identity

Q14

The Fourier series of a function in $(-\pi, \pi)$ is $f(x) = 1 + \sum_{n=1}^{\infty} \frac{2}{n} \sin nx$. Using Parseval's identity, find its RMS value.

Answer. Comparing, $\frac{a_0}{2} = 1 \Rightarrow a_0 = 2$, $a_n = 0$ and $b_n = \frac{2}{n}$. Since the RMS value is precisely Parseval's identity in the respective interval,

$$\begin{aligned} \bar{y}^2 &= \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} b_n^2 = \frac{4}{4} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{4}{n^2} = 1 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ &= 1 + 2 \left(\frac{\pi^2}{6} \right) \quad \left(\because \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \right) = 1 + \frac{\pi^2}{3} \end{aligned}$$

$$\bar{y} = \sqrt{1 + \frac{\pi^2}{3}} \approx 2.0712$$

Q15. Harmonic analysis – coefficients from a table

Q15

In harmonic analysis, how are a_0 , a_n and b_n obtained from N tabulated values of $y = f(x)$?

Answer. Each coefficient is twice the corresponding mean value:

$$a_0 = 2 \left(\frac{\sum y}{N} \right), \quad a_n = 2 \left(\frac{\sum y \cos nx}{N} \right), \quad b_n = 2 \left(\frac{\sum y \sin nx}{N} \right)$$

where N is the number of observations considered (the repeated end value is omitted).

Q16. Amplitude and phase of the first harmonic

Q16

In the harmonic analysis of an alternating current wave it is found that $a_1 = 0.363$ and $b_1 = 1.02$. Find the amplitude A_1 and the phase difference ϕ_1 of the first harmonic.

Answer.

$$A_1 = \sqrt{a_1^2 + b_1^2} = \sqrt{(0.363)^2 + (1.02)^2} = \sqrt{0.1318 + 1.0404} = \sqrt{1.1722}$$

$$\phi_1 = \tan^{-1} \left(\frac{b_1}{a_1} \right) = \tan^{-1} \left(\frac{1.02}{0.363} \right) = \tan^{-1}(2.810)$$

$$A_1 = 1.083, \quad \phi_1 = 1.23 \text{ radians } (\approx 70.4^\circ)$$

Q17. Sine coefficients of the sawtooth

Q17

Find b_n in the Fourier expansion of $f(x) = x$ in $(-\pi, \pi)$.

Answer. $f(x) = x$ is odd, so $a_0 = a_n = 0$ and

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi x \sin nx \, dx = \frac{2}{\pi} \left[-\frac{x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^\pi \\ &= \frac{2}{\pi} \left[-\frac{\pi \cos n\pi}{n} + 0 \right] = \frac{-2(-1)^n}{n} \end{aligned}$$

$$b_n = \frac{2(-1)^{n+1}}{n}$$

Q18. Half range series – when is it a pure cosine series?

Q18

What is a half range Fourier series? When does a function defined in $(0, \pi)$ possess a pure cosine series?

Answer. When $f(x)$ is defined over only the *half length* $(0, \pi)$ or $(0, \ell)$, its Fourier expansion contains either fully cosine terms or fully sine terms; such a series is called a **half range series**.

- Extending f as an **even** function to $(-\pi, \pi)$ gives a **half range cosine series** ($b_n = 0$).
- Extending f as an **odd** function to $(-\pi, \pi)$ gives a **half range sine series** ($a_0 = a_n = 0$).

Q19. Application – which harmonic resonates?**Q19**

A mass–spring–damper system with natural frequency $\omega_n = 5 \text{ rad/s}$ is driven by a square-wave force of period $2\pi \text{ s}$. Which harmonic of the input is resonant? Would a natural frequency of $\omega_n = 4 \text{ rad/s}$ be dangerous?

Answer. The fundamental frequency of the forcing is $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1 \text{ rad/s}$. A square wave contains only **odd** harmonics, i.e. $\omega = 1, 3, 5, 7, \dots \text{ rad/s}$. Resonance occurs when $n\omega_0 = \omega_n$:

$$n\omega_0 = 5 \Rightarrow n = 5 : \text{ the fifth harmonic is resonant}$$

If instead $\omega_n = 4$, then $n = 4$ would be required — but even harmonics are absent from a square wave, so **no harmonic is resonant** and the system is safe.

Q20. Application – phase lag at resonance**Q20**

For the system $m\ddot{x} + c\dot{x} + kx = F(t)$ with transfer function $G(s) = \frac{1}{ms^2 + cs + k}$, show that at resonance $\omega = \omega_n$ the steady-state response lags the force by exactly 90° , and give the gain there.

Answer. Putting $s = j\omega$,

$$G(j\omega) = \frac{1}{k - m\omega^2 + jc\omega}.$$

At $\omega = \omega_n = \sqrt{k/m}$ we have $k - m\omega_n^2 = 0$, so the real part vanishes:

$$G(j\omega_n) = \frac{1}{jc\omega_n} = \frac{-j}{c\omega_n}$$

which is purely imaginary and negative. Hence

$$|G(j\omega_n)| = \frac{1}{c\omega_n}, \quad \arg G(j\omega_n) = -90^\circ$$

That is, at resonance the response always lags the force by a quarter cycle whatever the damping, and its amplitude is governed by the damping constant c alone.

FOURIER SERIES ASSIGNMENT QUESTIONS

Assignment Questions

Instructions

Attempt all questions. Show every step of the working, state clearly whether the given function is even or odd before computing the coefficients, and quote the interval and its length 2π or 2ℓ at the start of each solution.

Assignment

- Find the Fourier series for the function

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ \sin x, & 0 < x < \pi \end{cases}$$

and hence evaluate

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \cdots = \frac{\pi - 2}{4}.$$

- Express $\sinh ax$ and $\cosh ax$ in Fourier series in the interval $-\pi < x < \pi$ of periodicity 2π .
- Find the Fourier series for the function

$$f(x) = \begin{cases} x, & 0 < x < \pi, \\ 2\pi - x, & \pi < x < 2\pi \end{cases}$$

and deduce that

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

- Express $f(x) = x \sin x$ in $0 < x < \pi$ as a Fourier cosine series and show that

$$1 + \frac{2}{1 \cdot 3} - \frac{2}{3 \cdot 5} + \frac{2}{5 \cdot 7} - \cdots = \frac{\pi}{2}.$$

- Find the Fourier series for the function

$$f(x) = \begin{cases} x, & -1 < x < 0, \\ x + 2, & 0 < x < 1 \end{cases}$$

and show that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

- Find the Fourier series for the function

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 1, \\ \frac{2-x}{2}, & 1 < x < 2. \end{cases}$$

7. Find the complex form of the Fourier series of the function $f(x) = e^x$ in $-\pi < x < \pi$.

8. The instantaneous current i at a time t of an alternating current wave is given by

$$i = I_1 \sin(\omega t + \alpha_1) + I_3 \sin(3\omega t + \alpha_3) + I_5 \sin(5\omega t + \alpha_5) + \dots$$

Find the effective value of the current.

9. The following table gives the variations of a periodic current over a period T :

t (sec)	0	$\frac{T}{6}$	$\frac{T}{3}$	$\frac{T}{2}$	$\frac{2T}{3}$	$\frac{5T}{6}$	T
i (amp)	1.98	1.30	1.05	1.30	-0.88	0.25	1.98

By numerical analysis show that there is a direct current part of 0.75 amp in the variable current. Obtain the amplitude of its first harmonic.

10. Find the Fourier series of $f(x)$ of periodicity 4, where

$$f(x) = \begin{cases} 0, & -2 < x < -1, \\ 1 + x, & -1 < x < 0, \\ 1 - x, & 0 < x < 1, \\ 0, & 1 < x < 2. \end{cases}$$

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